

Global Memory Market

CPU adds fuel to the 'higher and longer' upcycle thesis and 28E TAM at \$1.7trn; valuation framework transition in progress

- Memory demands broaden out from GPU to CPU.** CPU compute workload for AI is accelerating for orchestration (task coordination and logic), state management, and API execution. We have witnessed strong traction in Vera CPU adoption and expect major CSP customers to launch their own in-house CPU (e.g. Graviton/Axion) from 2H27 onwards. The CPU-to-GPU ratio has consistently trended up over the past three years (from 5.4:1 in '23 to 3.2:1 in '25) and we estimate it moving up to 2.4:1 by 28E (major reason behind 20-22% FY27E-28E server-grade DDR/LPDDR5 bit demand upward revision). We forecast AI CPU DRAM (headnode and standalone) demand at 11mn/17mn GB for FY27E-28E (19%/24% of total market demand). In the long term, Physical and World AI are the next areas of growth and a series of new applications (e.g. wearable and humanoid) are under development.
- Higher Memory TAM, S-D shortage likely to worsen next year, LTA remains to ASP stabilization.** Reflecting CoWoS model upward revisions ([note](#)), CPU memory upside, and CE demand degrade, we revise up our FY26E-28E memory TAM by 37-53% (vs [Mar-26 model](#)) and expect S-D shortage to worsen. This could theoretically add further upside risk to pricing for DRAM/NAND next year after a 220-250% y-y increase this year; however, we see a rather stable pricing trend supported by an increase in LTA contract sales mix. We estimate the Memory TAM will further expand to US\$1.3trn in 27E with ~50% market cap upside throughout next year using a 3.6x P/S multiple. Upside could be greater if we use the Market Cap to OP approach as memory stocks trade at a discount to their earnings (*Figures 2-4 for details*).
- HBM ASP upside risk from tighter S-D, higher wafer allocation trend continues.** We have introduced a new HBM S-D view with higher TAM (17-21% upward revision through 26E-28E) and the key changes are: i) ASIC demand acceleration (ASIC bit mix up from 33% in 26E to 39% in 27E), ii) NVDA Rubin GPU forecast cut in 26E offset by stronger GB build, iii) conservative HBM4E content assumption reflecting a longer 12Hi product life-cycle. Given the \$/bit crossover between HBM and non-HBM, we expect memory makers to turn vocal on raising HBM ASP next year (JPMe: +10% y-y on a like-for-like and +30% on a blended basis). As the HBM bit installation demand CAGR (+85% in the next three years) continues to outpace the rest of the market, we believe DRAM wafer capacity allocation to HBM will continue to rise (from 24% in 26E to 31% in 28E) adding greater pressure on conventional DRAM S-D.
- Memory gains substantial value within CSP capex, higher hardware spending to add confidence to memory earnings.** Memory is turning into a strategic asset ([MU CEO interview](#)) and we believe stable procurement is imperative for AI service operations for faster and high quality service quality. Memory as a % of total CSP hardware capex ranged between mid-teens % at the beginning of AI and is now anticipated to reach over half this year. Considering the importance of memory's role in AI compute systems, a higher value share is understandable, but many investors we have spoken to have

Technology - Semiconductors

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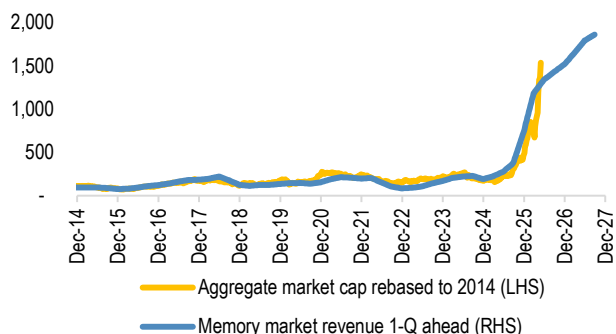
expressed concerns surrounding memory's further value share hike next year. This is one of the major reasons why memory stocks continue to trade at a discount to EPS. Higher CSP hardware capex ([JPM hardware research for details](#)) and greater longevity (more evidence on [NVIDIA's 2030E US\\$3-4trn spending outlook commentary](#)) should help alleviate concerns on memory earnings, in our view. Delayed utility infrastructure readiness or a slower new AI business model pickup could negatively impact hardware capex spending and eventually related memory demand.

- **Enterprise SSD leads NAND TAM expansion, capex upward revision not yet meaningful.** We see a clear trend in customers' preference for high-density and high-speed SSD with meaningful content growth (JPMc: over 40% GB per server increase y-y). We estimate the eSSD market to be over 500EB in 26E (43% of total NAND bit demand) and we forecast it to rise to over 1,100EB in the next two years at a +52% CAGR. Considering the ASP premium, eSSD value TAM could be over US\$300bn (larger than HBM TAM) in the next two years. On technology upfront, SLC based high speed SSD is gaining stronger traction on top of mainstream TLC SSD and we expect to see various solutions hitting the market from 2H26E (high I/OPs, HBF, and others). A growing SLC mix could be in favor of NAND S-D (higher bit per wafer trade-off vs TLC/QLC) as capex spending continues to remain below historical peak levels.
- **Capex acceleration to debottleneck the supply shortage.** Sharp memory economics improvement and a multi-year shortage outlook incentivize suppliers to accelerate capex spending (JPMc: US\$450bn in the next three years vs US\$300bn from [Dec-25 model](#)). **DRAM:** CY28-end WSPM at 2.8mn (880k increase from CY25-end) with three years accumulated capex of US\$364bn. 60% of incremental WSPM allocation to HBM and capex-to-sales average at 13% in the next three years. EUV procurement and infrastructure build out capacity is a key bottleneck. **NAND:** CY28-end WSPM at 1.44mn (165k increase from CY25-end) with US\$86bn spending (three year combined basis). Majority of spend is on tech migration and greenfield capacity impact materializing from 2H28. Relatively lower priority vs DRAM among top 3 producers, but we are beginning to see evidence of 2029 greenfield projects.
- **China Competition Assessment.** Volume share gains continue (DRAM share from 6% in '25 to 8-11% in 26E-28E and NAND share from 12% in '25 to 12-16% in 26E-28E) with above-industry capacity build out trends (DRAM/NAND capacity shares each at 18%/19% by CY28-end) and improving supply-chain localization. While all suppliers benefit from favorable S-D dynamics, inferior product mix likely limits the value share gain potential (DRAM/NAND value share at 10%/12% by CY28E). Key monitoring points are: 1) DRAM WSPM allocation amount to HBM (greater HBM wafer mix would result in tighter conventional DRAM S-D) and 2) Equipment export restriction policy update (i.e. MATCH Act - [link](#)).
- **Investment recommendations.** The global memory sector continues to outperform the respective index and ecosystem peers 2026-YTD and we maintain our multi-year bullish view on the sector. Fundamental key drivers remain: a) AI demand broadening out to CPU, b) rising HBM wafer allocation with greater trade loss ratio, and c) supply discipline in NAND (i.e. under-investment continuing) with potential supply constraints via SLC demand pickup. We acknowledge that the path to a new valuation framework will likely be a patchy one; however, we argue that AI has introduced a new spectrum of demand profiles and view a new valuation approach as necessary ([GMM: "LTA paves the path for a new valuation framework"](#) for details). Our top buy ideas include Korea Memory (**SKH** > **SEC** in the near-term), **Kioxia** in Japan, **Micron** in the U.S. We also like **Winbond** from Taiwan (covered by Jimmy Huang). For the broader supply chain in Asia, we like **TEL** (SPE) in Japan and **SIMO** (covered by Gokul Hariharan) in the controller solution space. Lastly, we view the risk-reward balance as unfavorable in selective stocks (**Hanmi**).

Key charts

Figure 1: Aggregate market cap vs leading memory market revenue trend (1Q ahead)

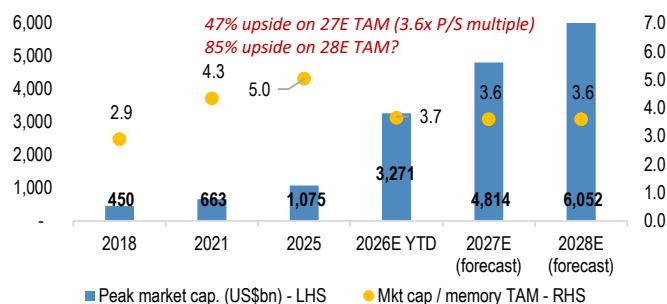
Rebased as of 1/1/2014, US\$bn-RHS



Source: WSTS, OMDIA, Bloomberg Finance L.P., J.P. Morgan estimates. Note: Market cap includes SEC, SK Hynix, Micron, Nanya Technology, Western Digital. Kioxia / SNDK added from 1Q25 and excluded WDC.

Figure 2: Memory TAM vs peak mkt cap/TAM analysis

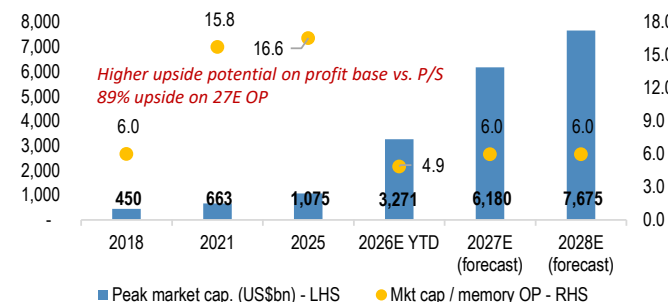
US\$bn, peak mkt cap/TAM (x)



Source: Bloomberg Finance L.P., WSTS, Company data, J.P. Morgan estimates. Note: only considered SEC, SKH, and MU market cap.

Figure 3: Memory TAM vs peak mkt cap/memory OP analysis

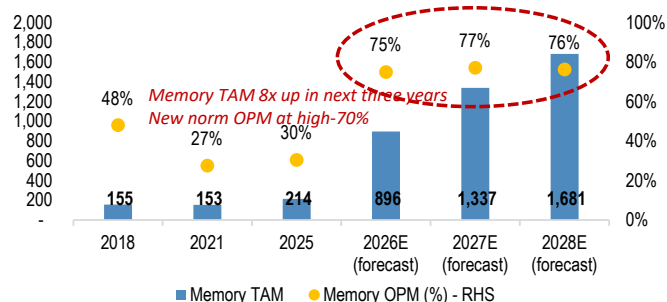
US\$bn, peak mkt cap/memory EBIT (x)



Source: Bloomberg Finance L.P., WSTS, Company data, J.P. Morgan estimates. Note: only considered SEC, SKH, and MU market cap.

Figure 4: Memory TAM vs Memory OPM

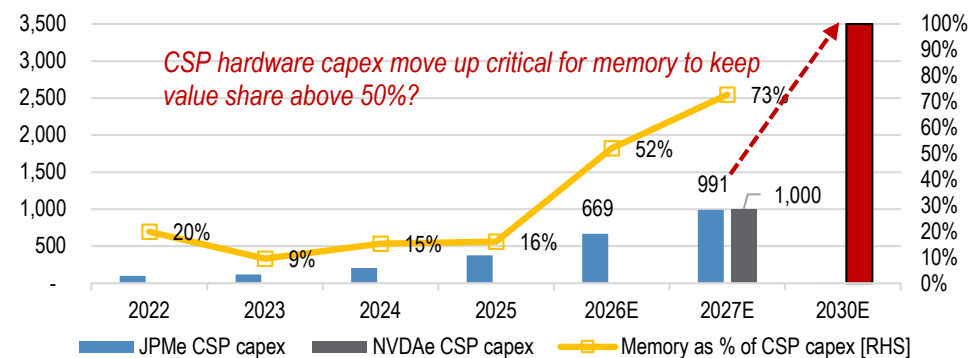
US\$bn, % - RHS



Source: Bloomberg Finance L.P., WSTS, Company data, J.P. Morgan estimates. Note: only considered SEC, SKH, and MU market cap.

Figure 5: CSP capex and AI memory % share

US\$bn, % - RHS



Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates.

Table 1: Market cap changes across AI ecosystem partners

US\$bn, %

US\$bn	Top 4 CSP	AI semi	Foundry	Memory	AI ecosystem mkt cap.	SOX	NASDAQ index	DOW Jones index
1Q24	8,122	3,226	632	595	12,575	5,795	39,807	5,254
2Q24	8,871	4,109	772	623	14,376	6,991	39,119	5,460
3Q24	8,652	4,113	784	491	14,039	6,900	42,330	5,762
4Q24	9,243	4,667	850	395	15,156	7,306	42,544	5,882
1Q25	8,162	3,651	711	423	12,947	6,040	42,002	5,612
2Q25	10,027	5,448	940	557	16,972	8,481	44,095	6,205
3Q25	10,979	6,427	1,111	721	19,238	10,022	46,398	6,688
4Q25	11,508	6,594	1,280	1,145	20,526	10,608	48,063	6,846
1Q26	9,907	6,122	1,424	1,411	18,863	10,738	46,342	6,529
2Q26YTD	12,435	8,233	1,948	3,369	25,986	15,469	50,669	7,564

% changes

2Q24	9%	27%	22%	5%	14%	21%	-2%	4%
3Q24	-2%	0%	1%	-21%	-2%	-1%	8%	6%
4Q24	7%	13%	8%	-19%	8%	6%	1%	2%
1Q25	-12%	-22%	-16%	7%	-15%	-17%	-1%	-5%
2Q25	23%	49%	32%	31%	31%	40%	5%	11%
3Q25	9%	18%	18%	30%	13%	18%	5%	8%
4Q25	5%	3%	15%	59%	7%	6%	4%	2%
1Q26	-14%	-7%	11%	23%	-8%	1%	-4%	-5%
2Q26YTD	26%	34%	37%	139%	38%	44%	9%	16%

mkt cap. mix within the AI ecosystem

1Q24	65%	26%	5%	5%	100%	
2Q24	62%	29%	5%	4%	100%	
3Q24	62%	29%	6%	3%	100%	
4Q24	61%	31%	6%	3%	100%	
1Q25	63%	28%	5%	3%	100%	
2Q25	59%	32%	6%	3%	100%	
3Q25	57%	33%	6%	4%	100%	
4Q25	56%	32%	6%	6%	100%	
1Q26	53%	32%	8%	7%	100%	
2Q26YTD	48%	32%	7%	13%	100%	

Source: Bloomberg Finance L.P., Note: 1) Top 4 CSP includes GOOG/MSFT/AMZN/META;; 2) AI semi includes NVDA/AMD/AVGO/MRVL; 3) Foundry includes TSMC; 4) Memory includes SKH/MU/SEC; 5) NASDAQ and DJIS in index vs others in absolute market cap; 6) Market cap. based on qtr-end; 7) colors indicate % performance rank within the qtr.

Table 2: OPM comparison across AI ecosystem partners

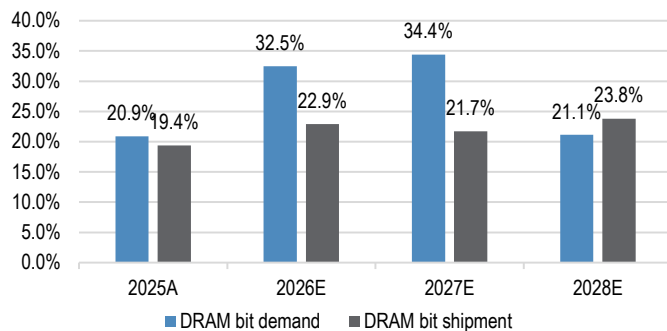
%

OPM comparison	2022	2023	2024	2025	2026E	2027E	2028E
CSP Cloud	29%	29%	35%	36%	37%	38%	39%
AI Semi	39%	49%	57%	57%	63%	64%	63%
Memory	23%	-30%	26%	36%	76%	75%	72%
Foundry	46%	40%	42%	49%	56%	55%	56%

Source: Company data, J.P. Morgan estimates, Bloomberg Finance L.P. *note: Memory OPM includes both DRAM and NAND.

Figure 6: DRAM bit demand vs bit supply

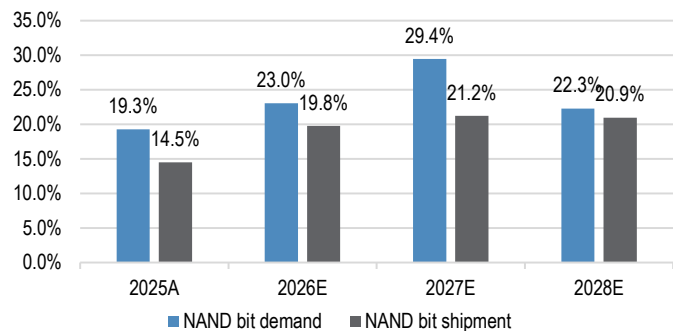
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Source: J.P. Morgan estimates.

Figure 7: NAND bit demand vs bit supply

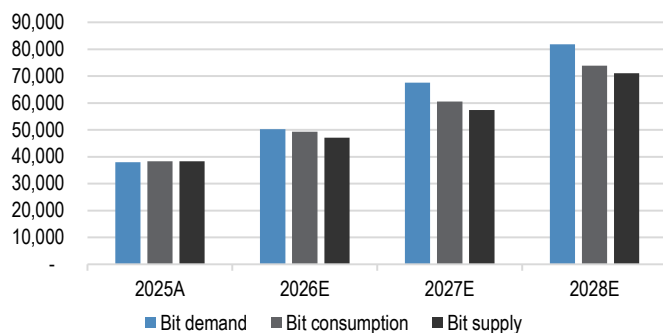
%



Source: J.P. Morgan estimates.

Figure 8: DRAM bit demand vs bit consumption vs bit supply in absolute terms

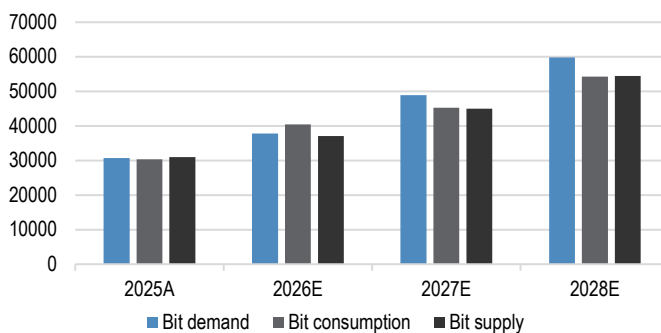
8Gb. Eq. (M)



Source: J.P. Morgan estimates.

Figure 9: NAND bit demand vs bit consumption vs bit supply in absolute terms

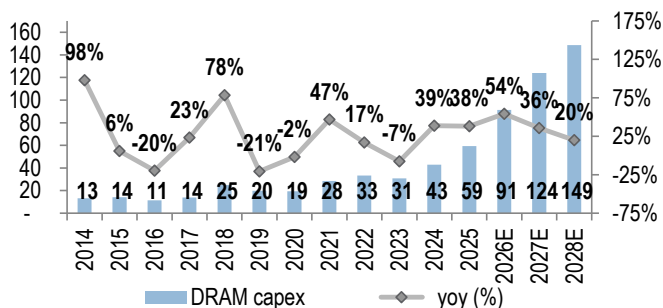
256Gb Equi in Mn



Source: J.P. Morgan estimates.

Figure 10: DRAM capex and Y/Y trend

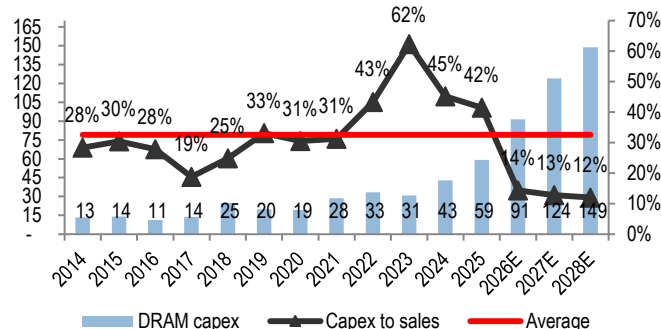
US\$bn, %



Source: Company data, J.P. Morgan estimates.

Figure 11: DRAM capital intensity

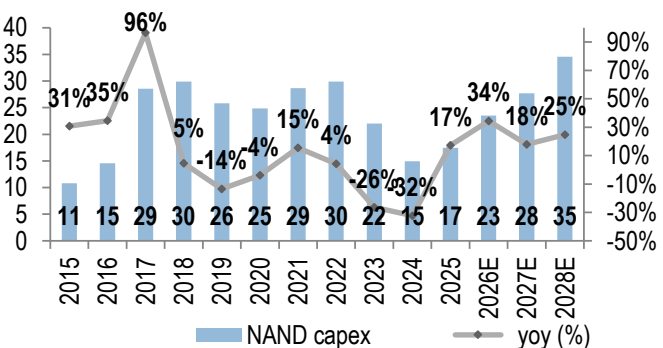
US\$bn, %



Source: Company data, J.P. Morgan estimates.

Figure 12: NAND capex and Y/Y trend

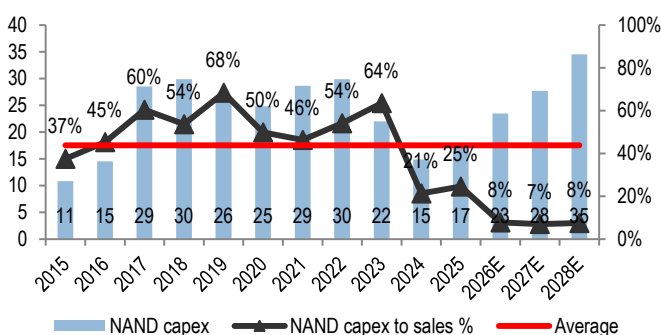
US\$bn, %



Source: Company data, J.P. Morgan estimates.

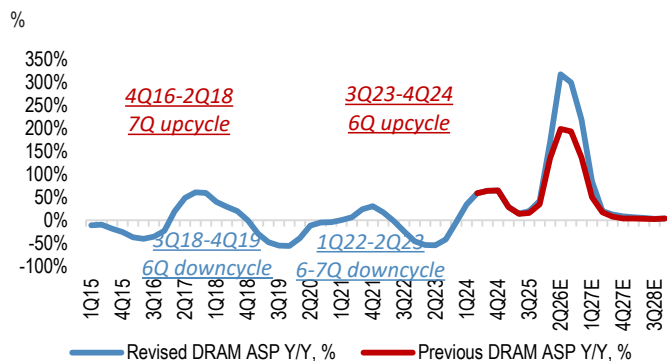
Figure 13: NAND capital intensity

US\$bn, %



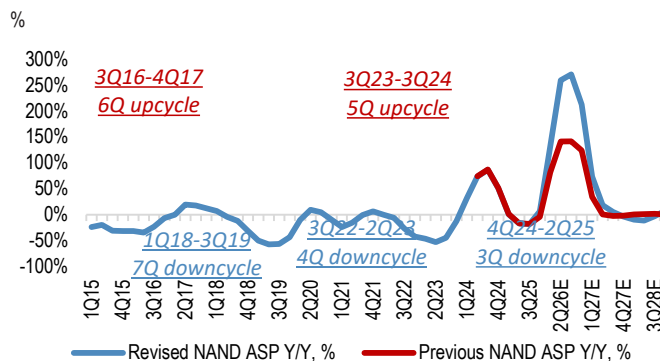
Source: Company data, J.P. Morgan estimates. Note: Average from 2011-25.

Figure 14: DRAM ASP Y/Y% change vs previous assumptions



Source: WSTS, J.P. Morgan estimates.

Figure 15: NAND ASP Y/Y% change vs previous assumptions



Source: Gartner®, J.P. Morgan estimates.

Figure 16: Memory Semiconductor cycle scenario analysis

Memory Semiconductor cycle scenario

Bull cycle (probability = 25%)

- **LLM foundation model:** parameter expansion continues to expand at a fast pace, sharp model transition acceleration into physical/world AI
- **Memory consumption architecture:** limited software stack improvement, hardware intensity remains at elevated level in every new computing architecture
- **CSP hardware spending:** token consumption curve remains exponential followed by sharp AI agent monetization enabling strong capex increase
- **Consumer demand:** resilient demand support after near-term unit sales correction, edge AI demand picks up amidst component cost inflation
- **Multi-year supply agreement:** suppliers and customers able to justify spending to each other with high investment and order visibility
- **WFE spending and competition:** aggressive infrastructure capex but reasonable equipment spending and China fails to scale-up/out due to restriction
- **New technology development:** customers actively deploying new form-factors (HBF, LLW, and others) and wafer-to-die penalty ratio sharply worsens

Base case cycle (probability = 60%)

- **LLM foundation model:** parameter continues at moderate pace, gradual model transition acceleration into physical/world AI
- **Memory consumption architecture:** continued memory intensity albeit software stack improvement, hardware intensity continues to grow in new architecture
- **CSP hardware spending:** token consumption curve remains robust and moderate AI agent monetization enabling stable capex increase
- **Consumer demand:** mid/low-end segment continues to shrink gradually after near-term correction, edge AI demand slowly progressing
- **Multi-year supply agreement:** suppliers and customers reach to consensus on investment scale with certain bindings and reasonable order visibility
- **WFE spending and competition:** aggressive infrastructure capex and ongoing equipment spending and China continues to face scale-up/out challenges
- **New technology development:** customers selectively adopt new form-factors and wafer-to-die penalty ratio gradually worsens

Bear case cycle (probability = 15%)

- **LLM foundation model:** parameter expansion takes a pause, slow model transition acceleration into physical/world AI at fast pace, and rising funding concerns
- **Memory consumption architecture:** software stack breakthrough, hardware intensity flattens due to BOM cost pressure in new architecture
- **CSP hardware spending:** token consumption curve remains linear and slow AI agent monetization translating into muted capex spending
- **Consumer demand:** demand destruction across all segments and edge AI project delayed by multiple years
- **Multi-year supply agreement:** suppliers and customers fail to comply to agreement with limited order visibility and surplus capacity turns into over-supply
- **WFE spending and competition:** aggressive infrastructure and equipment capex and China shows meaningful pickup in bit/wafer productivity and product quality
- **New technology development:** customers lose appetite to try new form-factors and wafer-to-die penalty concern dissipates

Source: J.P. Morgan estimates.

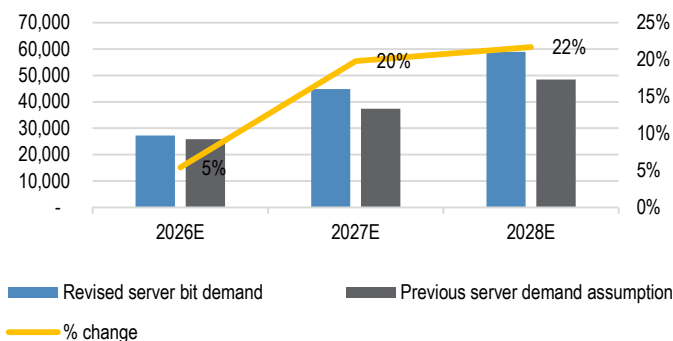
CPU: Pouring more demand

Following the GPU-dense AI narrative, the biggest surprise in 1Q26 earnings was the CPU demand upside serving as the orchestration layer and the critical control plane for AGI: 1) NVDA guided to US\$20bn standalone Vera CPU sales in 2026 ([note](#)); 2) AMD revised up the server CPU TAM to grow 35%+ per annum to \$120bn by 2030 (vs prev.: 18%) ([note](#)); 3) INTC guided the CPU:GPU ratio moving from ~1:8 to 1:4 and hinted eventual parity thanks to the increasing compute requirements of agentic AI ([note](#)).

From a memory industry perspective, we believe a few positive developments that stand out are: 1) added memory demand driver from NVDA entering the CPU market where our team estimates 600K/3M Vera CPU unit in 26/27E ([report](#)); 2) strong upside to AI server bit demand not only from increased AI server racks but also growing CPU:GPU ratio especially in ASIC server racks. Overall, we raise server bit demand assumption by 5%-22% through 26E-28E and emphasize that 60%+ of the upward revision in 27-28E originates from AI server (AI headnode + AI standalone CPU) demand upside. We estimate AI server bit demand (AI headnode + AI CPU, excluding HBM) accounts for ~30%+ of total DRAM bit demand in 27-28E (vs previously : 14%).

Figure 17: Server bit demand forecast changes

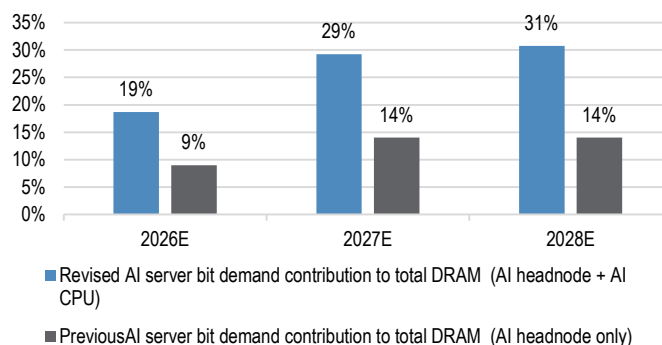
8Gb equiv. Mn, % - RHS



Source: J.P. Morgan estimates.

Figure 18: AI server bit demand contribution changes

%



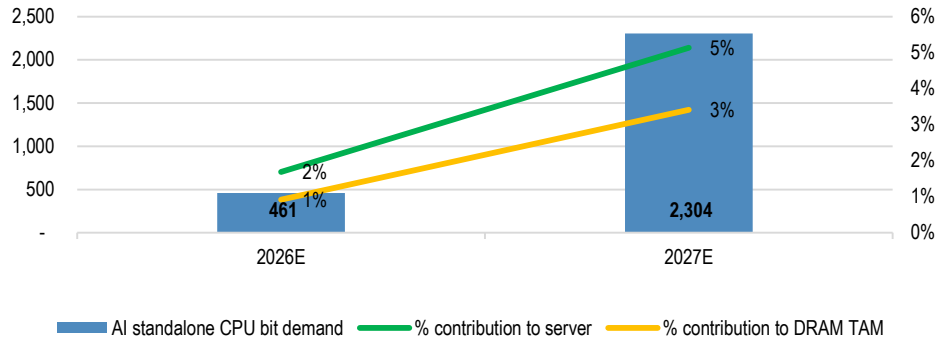
Source: Company data, J.P. Morgan estimates.

Calibrating the demand upside from standalone Vera CPU

Based on our supply-chain checks, we believe the Vera CPU comes in two memory configurations (768GB content based on 4 modules of 192GB SOCAMM2 and 1.5TB content based on 8 modules of 192GB SOCAMM2). Since a typical two-socket server carries ~1.5TB of memory content (or ~754GB of memory content per CPU), we take the low-end of the content configuration and assume the incremental demand from AI CPUs to reach 461M / 2.3bn GB of bit demand in 26E/27E (1%/3% of total memory demand in 26/27E) based on 600K/3M Vera CPU units in 26/27E. Albeit the contribution looks miniscule from a total contribution perspective, we envision the advent of AGI will create a strong demand tailwind for CPU and eventually for memory amidst multiple other fabless companies entering this market including ARM's AGI CPU ([report](#)) and QCOM's ARM-based server CPU ([report](#)). Nevertheless, the basis of our cautious AI CPU demand assumption comes from: 1) limitations of CoWoS-R capacity that may potentially cap Vera CPU production in the near-term; 2) more evidence of AI CPU production from other players like QCOM/ARM in the supplychain; and 3) limited memory capacity to fulfill overwhelming demand.

Table 3: AI standalone CPU bit demand and % contribution to server/DRAM TAM

8Gb Equiv. Mn, % - RHS



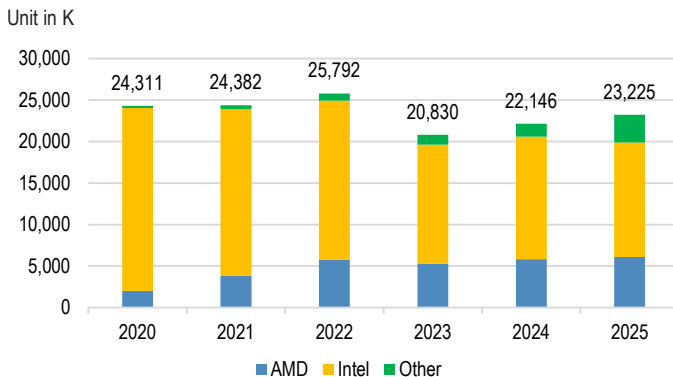
Source: J.P. Morgan estimates.

Memory density map for CPU and server

To simplify, the memory capacity for a CPU is calculated by multiplying the **number of channels * DIMM per channel * capacity per DIMM**. Depending on the CPU slot configuration by server (either one, two, or four sockets per server), the system memory capacity is derived by multiplying 1x - 4x to the memory capacity per CPU. For instance, the memory capacity for a CPU carrying 12 memory channels, 2 DPC (DIMM Per Channel), and 64GB DIMM would be 1.5TB (12 * 2 * 64). Then, if the server rack carries a two socket specification (two physical slots to install two independent CPU chips onto a single motherboard), the memory capacity per server system would be 3TB (1.5TB * 2).

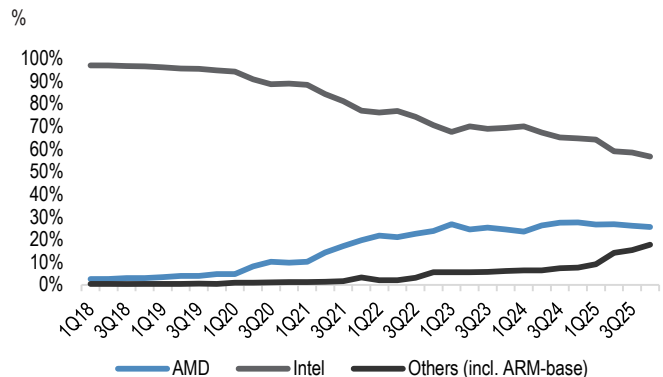
Based on Gartner® data, total server CPU unit TAM reached 23M in 25A with two sockets per server configuration accounting for ~80%, translating to roughly 1.85 CPU sockets per server.

Figure 19: Server CPU unit TAM



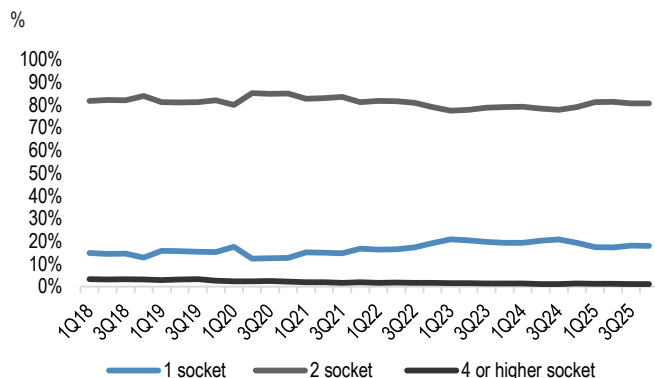
Source: Gartner®

Figure 20: Server CPU market share



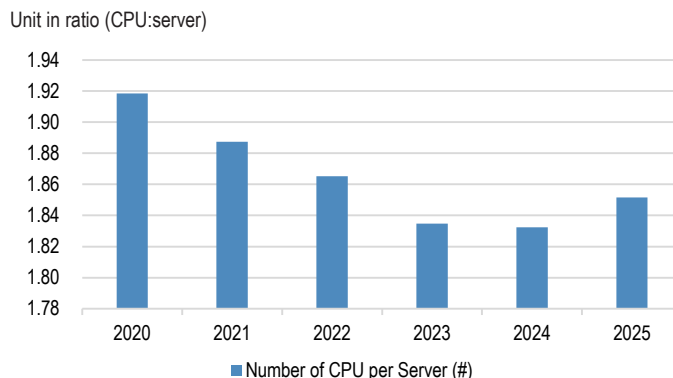
Source: Gartner®

Figure 21: Server CPU configuration



Source: Gartner®

Figure 22: CPU per server trend



Source: Gartner®

Implications for memory from a narrowing GPU-to-CPU ratio

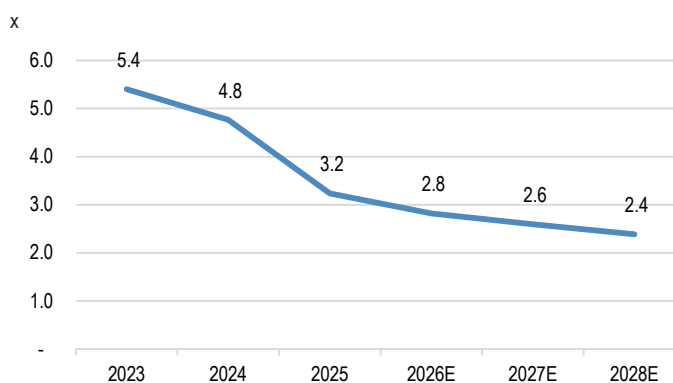
As the CPU plays a critical role in the orchestration layer for AGI, we expect a narrowing GPU-to-CPU ratio to present meaningful demand upside to memory (mainly for ASIC, in our view). Based on our bottom-up analysis, we forecast the GPU-to-CPU ratio to fall from ~5 in 23A to ~2 :1 in 28E leading us to raise our CPU system memory content assumption. We expect this trend to be more notable in ASIC racks (more CPU density per rack per GPU) as NVDA racks are already adopting a 2:1 GPU-to-CPU ratio.

Figure 23: GPU to CPU assumption for AI headnode demand

Unit in K, x	2025	2026E	2027E	2028E
Unit 000s				
GPU/ASIC	11,216	16,885	24,118	28,500
NVDA	6,200	8,920	9,900	10,750
AMD	675	550	978	1,100
ASIC/others	4,341	7,415	13,240	16,650
AI grade CPU	3,469	5,982	9,289	11,941
NVDA	2,720	4,186	4,394	5,314
AMD	169	138	245	275
ASIC/others	580	1,658	4,650	6,352
GPU to CPU (x)	3.2	2.8	2.6	2.4
NVDA	2.3	2.1	2.3	2.0
AMD	4.0	4.0	4.0	4.0
ASIC/others	7.5	4.5	2.8	2.6

Source: J.P. Morgan estimates.

Figure 24: GPU to CPU assumption

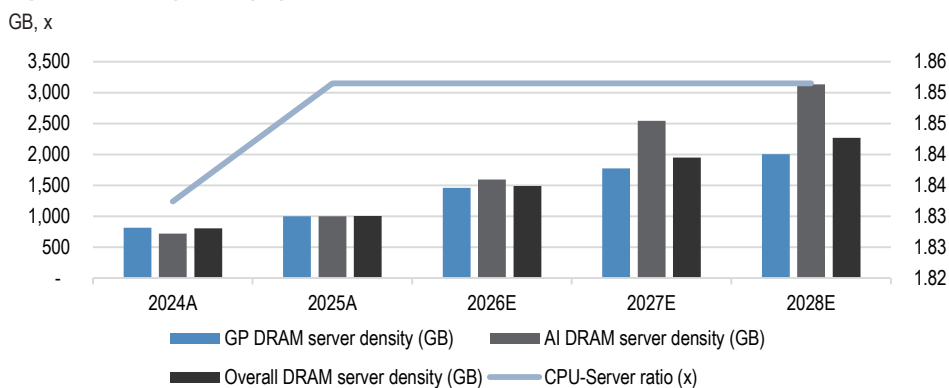


Source: J.P. Morgan estimates.

Memory density map for server CPU

Looking at server CPU specs for in-house custom silicons (i.e. AWS Graviton 5, MSFT Cobalt 100, and GOOG Axion), we noticed 1 DPC memory configuration (compared to 1-2 DPC configuration for others) which theoretically halves the memory content per CPU. Nevertheless, we expect higher CPU slots per server should more than offset the content shortfall per CPU (see the memory density map in Table 4 for more details).
<where?>

Figure 25: Memory Density by Server Applications



Source: Company data, J.P. Morgan estimates.

Figure 26: Average DRAM capacity per CPU

Average DRAM capacity per CPU (per one socket)	Mainstream Server DIMM				
	32GB	64GB	96GB	128GB	256GB
Number of Channel = 8 DIMM per CPU = 1	256	512	768	1,024	2,048
Number of Channel = 8 DIMM per CPU = 2	512	1,024	1,536	2,048	4,096
Number of Channel = 12 DIMM per CPU = 1	384	768	1,152	1,536	3,072
Number of Channel = 12 DIMM per CPU = 2	768	1,536	2,304	3,072	6,144

Source: J.P. Morgan estimates.

Figure 27: Average DRAM capacity per server

Average DRAM capacity per server (1.85 socket per server)	Mainstream Server DIMM				
	32GB	64GB	96GB	128GB	256GB
Number of Channel = 8 DIMM per CPU = 1	474	948	1,422	1,896	3,792
Number of Channel = 8 DIMM per CPU = 2	948	1,896	2,844	3,792	7,584
Number of Channel = 12 DIMM per CPU = 1	711	1,422	2,133	2,844	5,688
Number of Channel = 12 DIMM per CPU = 2	1,422	2,844	4,266	5,688	11,376

Source: J.P. Morgan estimates. Note: Assume 1.85 CPU sockets per server

Table 4: Server CPU spec comparison

Brand	Code	Architecture	Process node	Release	vCPUs	Memory density per socket	# of memory channel support	DIMM per Channel	PCIe Gen
Intel	Ice Lake	x86	10nm	1H21	Up to 40 P-cores per socket	Up to 4TB per socket	Up to 8 channels of DDR4-3200		1-2 PCIe Gen 4 (Lane # = 64)
Intel	Sapphire Rapids	x86	10nm	1H23	Up to 64 P-cores per socket	Up to 4TB per socket	Up to 8 channels of DDR5-4800		1-2 PCIe Gen 5 (Lane # = 80)
Intel	Emerald Rapids	x86	7nm	2H23	Up to 64 P-cores per socket	Up to 4TB per socket	Up to 12 channels of DDR5-5600/6400		1-2 PCIe Gen 5 (Lane # = 80)
Intel	Sierra Forest	x86	3nm	1H24	Up to 288 E-cores per socket	Up to 1-2TB+ per socket	Up to 12 channels of DDR5-5600/6400		1-2 PCIe Gen 5 (Lane # = 136)
Intel	Granite Rapids	x86	3nm	2H24	Up to 128 P-cores per socket	Up to 3TB per socket or more	Up to 12 channels of DDR5-5200/6400		1-2 PCIe Gen 5 (Lane # = 136)
Intel	Cleanwater Forest	x86	2nm	1H26	Up to 288 E-cores per socket	Up to 3TB per socket or more	Up to 12 channels of DDR5-8000		1-2 PCIe Gen 5 (Lane # = 96)
Intel	Diamond Rapids	x86	2nm	2H27	Up to 256-512 P-cores per socket	Up to 8TB per socket	Up to 16 channels of future DDR solutions		1-2 PCIe Gen 6 (Lane # = n/a)
Intel	Coral Rapids	x86	TBD	2H28	n/a	n/a	Up to 16 channels of future DDR solutions		n/a n/a
AMD	Milan	Zen 3	5nm	1H21	Up to 64 cores per socket	Up to 6TB per socket	Up to 12 channels per DDR4-xxxx		1-2 PCIe Gen 5 (Lane # = 128)
AMD	Genoa	Zen 4	5nm	2H22	Up to 96 cores per socket	Up to 6TB per socket	Up to 12 channels per DDR5-xxxx		1-2 PCIe Gen 5 (Lane # = 128)
AMD	Bergamo	Zen 4c	5nm	1H23	Up to 128 cores per socket	Up to 6TB per socket	Up to 12 channels of DDR5-4800		1-2 PCIe Gen 5 (Lane # = 128)
AMD	Siena	Zen 4c	5nm	2H23	Up to 64 cores per socket	Up to 1.15TB per socket	Up to 6 channels of DDR5-4800		1-2 PCIe Gen 5 (Lane # = 96)
AMD	Turin	Zen 5/5c	3nm/4nm	2H24	Up to 192 cores per socket	Up to 6TB per socket	Up to 12 channels of DDR5		1-2 PCIe Gen 5 (Lane # = 128)
AMD	Venice	Zen 6	2nm	2026	Up to 256 cores per socket	Up to 8TB per socket	Up to 16 channels of future DDR solutions		1-2 PCIe Gen 6 (Lane # = 128)
AMD	Verano	Zen 7	1.6nm	2027	n/a	n/a	Up to 8 channels of future DDR solutions or LPDDR5X SOCAMM2		n/a PCIe Gen 6 (Lane # = n/a)
NVDA	Grace	ARM v9	4nm	2024	Up to 72 core / socket	Up to 120GB-480GB per each CPU	Up to 32 channels		n/a PCIe Gen 5 (Lane # = 128)
NVDA	Vera	ARM v9.2	3nm	2026	Up to 88 core / socket	Up to 1.5TB	n/a		n/a PCIe Gen 6 (Lane # = 16)
AWS	Graviton 3	ARM v8.4	5nm	2022	Up to 64 core / socket	Up to 512 GiB on 64-core R7g	Up to 8 channels of DDR5-4800 memory		1-2 PCIe Gen 5 (Lane # = 64)
AWS	Graviton 4	ARM v9	5nm	2024	Up to 96 core / socket	Up to 3 TiB of RAM (= 3.2TB)	Up to 12 channels of DDR5-5600 memory		1-2 PCIe Gen 5 (Lane # = 96)
AWS	Graviton 5	ARM v9	3nm	2026	Up to 192 core / socket	Up to 3TB	Up to 12 channels of DDR5-7200, supporting future upgrade to 8800 MT/s		1 PCIe Gen 6 (Lane # = 96)
Ampere	Ampere One	ARM v9	5nm	2026	Up to 192 core / socket	Up to 4TB	Up to 8 channels of DDR5-5200		1 PCIe Gen 5 (Lane # = 128)
Ampere	Ampere One-M	ARM v9	5nm	2026	Up to 192 core / socket	Up to 1.5TB	Up to 12 channels of DDR5-5600		1 PCIe Gen 5 (Lane # = 96)
MSFT	Cobalt 100	ARM v9	5nm	2024	Up to 128 core / socket	Up to 672 GiB of RAM (=720GB)	Up to 12 channels of DDR5 memory support		1 PCIe Gen 5 (Lane # = 96)
Google	Axion	ARM v9	3nm	2025	Up to 72 core / socket	Up to 768GB	Up to 8 channels of DDR5 memory support		1 PCIe Gen 5 (Lane # = 96)

Source: Tech Powerup.

Server bit demand sensitivity analysis based on content assumption changes

Given the multiple variants of CPU content per server rack, we have provided a bull-case sensitivity analysis assuming the below.

- Higher AI headnode-related CPU unit demand by lowering the GPU to CPU ratio to 2.7 and 2.3 in 26/27E (vs base case: 2.8 and 2.6);
- Higher Vera CPU standalone production of 700K/3.5M in 26/27E (vs base case: 600K/3M);
- Higher DRAM content per standalone vera CPU of 1.2TB/1.3TB (vs base case: 768GB in both years);
- Accelerating GP server unit growth by 22%/20% y-y growth in 26/27E (vs base case: 20%/16% growth) and content growth acceleration to 50%/27% growth in 26/27E (vs base case: 46%/22%)

Assuming the above, we see 6%/20% server bit demand upside in 26/27E, and we believe the bull case demand assumptions are mostly feasible with the key bottleneck being memory supply which will ironically limit the total DRAM TAM. Historically, a mismatch in S/D dynamics impacted pricing in the event of memory shortage, but with memory margins reaching near 80%+, we expect memory makers to slow ASP hikes to soften demand destruction and seek LTA and higher demand/order visibility.

Table 5: Server bit demand sensitivity analysis

Type	DRAM	2025	2026E	2027E	2026E (bull)	2027E (bull)
GP	DDR/LPDDR (mn GB)	10,567	18,469	26,023	19,292	29,277
	Y-Y changes (%)	20%	75%	41%	83%	52%
	Server (000s)	10,571	12,646	14,660	12,857	15,419
	Y-Y changes (%)	-2%	20%	16%	22%	20%
	Density (GB)	1,000	1,461	1,775	1,500	1,899
	Y-Y changes (%)	23%	46%	22%	50%	27%
AI	AI headnode demand (mn GB)	1,713	4,644	10,809	4,999	12,953
	Y-Y changes (%)		171%	133%	192%	159%
	AI grade CPU (K)	3,469	5,982	9,289	6,317	10,502
	Y-Y changes (%)		72%	55%	82%	66%
	AI GPU/ASIC unit (K)	11,216	16,885	24,118	16,885	24,118
	GPU to CPU (x)	3.2	2.8	2.6	2.7	2.3
	DRAM content per AI CPU (GB)	494	776	1,164	791	1,233
	Y-Y changes (%)		57%	50%	60%	56%
	Standalone CPU demand (mn GB)		461	2,304	806	4,704
	Y-Y changes (%)			400%	75%	483%
	Vera CPU (K)		600	3,000	700	3,500
	Y-Y changes (%)			400%		400%
	DRAM content per AI CPU (GB)		768	768	1,152	1,344
	Y-Y changes (%)			0%	50%	17%
Sum	Server total (mn GB)	12,280	23,574	39,136	25,097	46,934
	Y-Y changes (%)		92%	66%	104%	87%
	Difference vs. base (%)				6%	20%

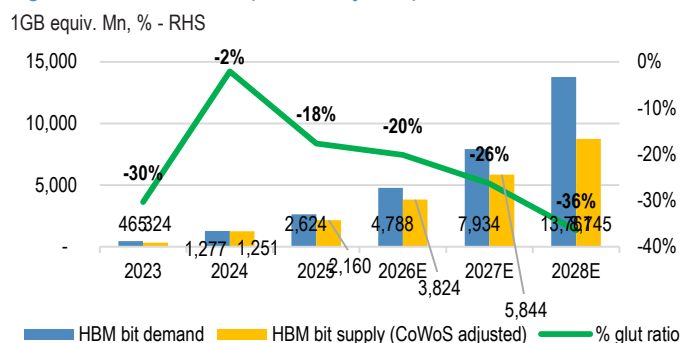
Source: J.P. Morgan estimates.

HBM update

HBM S/D update: shortage continues to persist

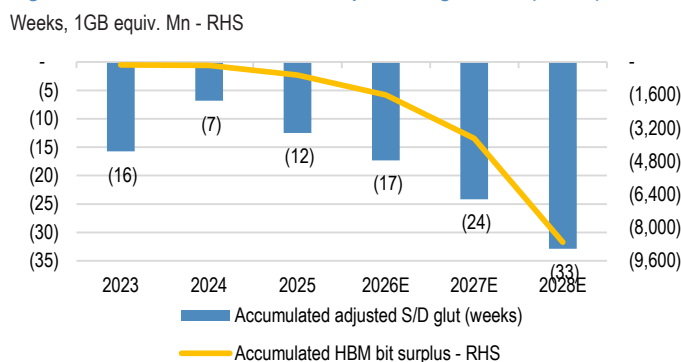
Following the CoWoS model update ([report](#)), we update our HBM S-D view where key changes are: 1) stronger HBM ASP in 27E from resilient HBM3E/ASIC demand (especially from GOOG/AMZN); 2) downward revision to NVDA Rubin GPU forecast in 26E from HBM4 sourcing uncertainty offset by Blackwell GPU(GB300) unit upside while assuming higher Rubin volume mix in 27E; and 3) conservative HBM4E content assumption for Rubin Ultra (revised down from having only 1TB SKU to include both 12 Hi and 16 Hi). We raise HBM TAM by 17%-21% through 26-28E and expect HBM S-D to remain in a DD% shortage throughout 28E.

Figure 28: HBM S/D ratio (CoWoS adjusted)



Source: Company data, J.P. Morgan estimates.

Figure 29: Accumulated HBM bit surplus and glut ratio (weeks)



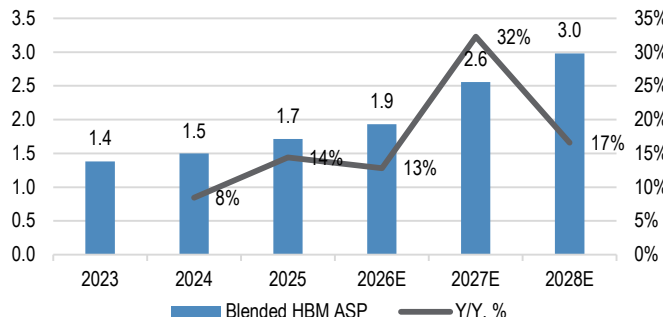
Source: Company data, J.P. Morgan estimates.

HBM ASP

Contrary to our previous expectations of HBM3E ASP declining 2%/flat in FY26E/27E, respectively, the strong demand environment across both GPU and ASICs with the majority of the ASIC volumes adopting HBM3E by 27E, we expect HBM3E ASP to be flat/19% in FY26E/27E as tight HBM3E S-D creates a favorable market environment for memory makers. Consequently, we continue to see blended HBM ASP rising in 26E by mid-teens% aided by a 29% HBM4 pricing premium. Interestingly, despite some HBM capacity coming online in 2027, we expect blended HBM ASP to rise by a record 32% Y/Y driven by 19%/15% growth in HBM3E and HBM4 ASP, respectively, with HBM4E being priced at a 61% premium to HBM3E due to severe tightness in the supply chain. Overall, we foresee a tight HBM S-D market throughout 2028 and expect HBM TAM growth rate to accelerate in FY27E thanks to Rubin Ultra ramp as well as ASIC content upgrade to HBM3E from HBM.

Figure 30: HBM ASP and Y/Y% trend

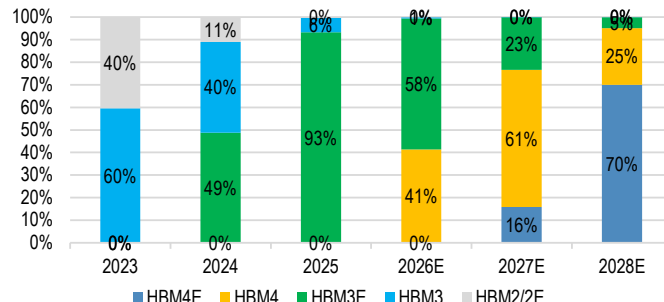
US\$/Gb, % - RHS



Source: Company data, J.P. Morgan estimates.

Figure 31: HBM sales mix by product category

%



Source: Company data, J.P. Morgan estimates.

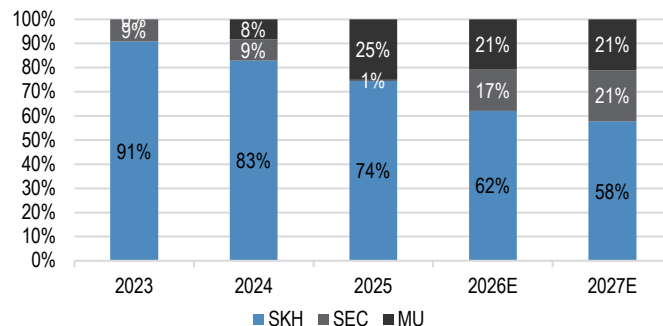
HBM wallet share update

We tick up Samsung HBM4 market share within NVDA to high-teens% in 26E (HBM4 related wallet share to be higher of ~31%) to reflect the current HBM4 qualification progress where we believe Samsung is ahead vs competitors. We attribute the strength to SEC's 1nm mono-die (SKH/MU relies on 1nm) and SF's 4nm base die, and gradually improving back-end yield. On the ASIC front, we expect the share trend to remain broadly unchanged with minor share gains from MU given its aggressive HBM capacity expansion plans until 27E (US ID1 fab and Singapore packaging fab). The qualification of HBM4 has altered the wallet share dynamics, and we anticipate that the introduction of HBM4E will trigger a new wave of competition among memory suppliers, with companies likely to compete on factors such as pin speed, logic die sourcing, and yield performance.

As expected from the earnings conference, Samsung began shipping samples of 12-Hi 48GB HBM4E for qualification ([link](#)). We expect rack roll-out adopting HBM4E to be much later (JPMe: 2H27E) as SEC also gudies plans to expand the lineup including 8-Hi HBM4E (32GB) up to 16-Hi HBM4E (64GB). While the sample shipment does not indicate technology leadership, we see strong improvement in Samsung's HBM technology capability.

Figure 32: HBM wallet share within NVDA

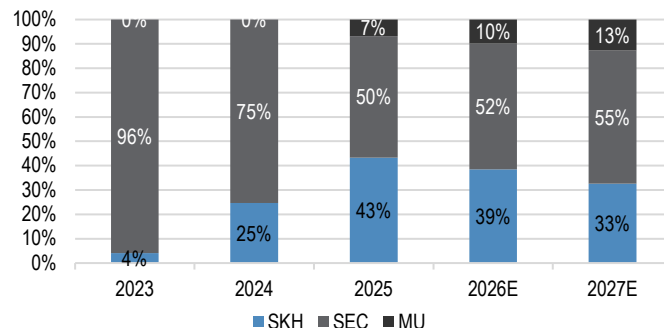
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Source: Company data, J.P. Morgan estimates.

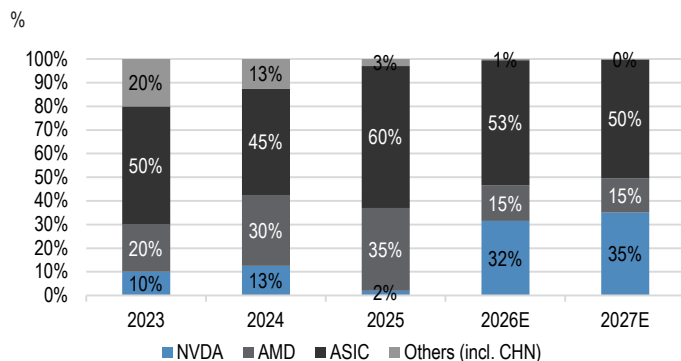
Figure 33: HBM wallet share within ASIC

%



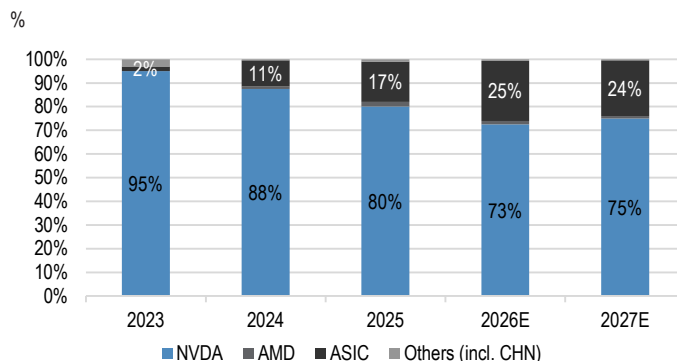
Source: Company data, J.P. Morgan estimates.

Figure 34: SEC - HBM sales mix by customers



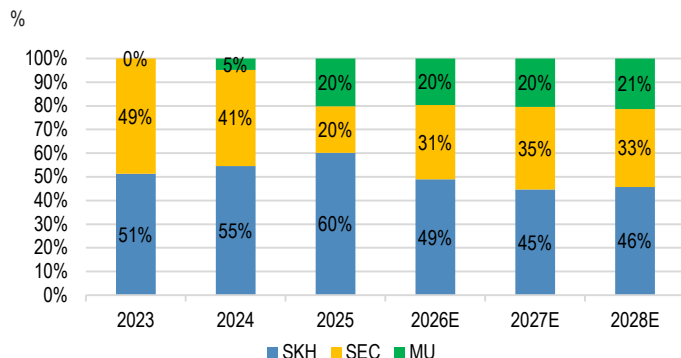
Source: Company data, J.P. Morgan estimates.

Figure 35: SKH - HBM sales mix by customers



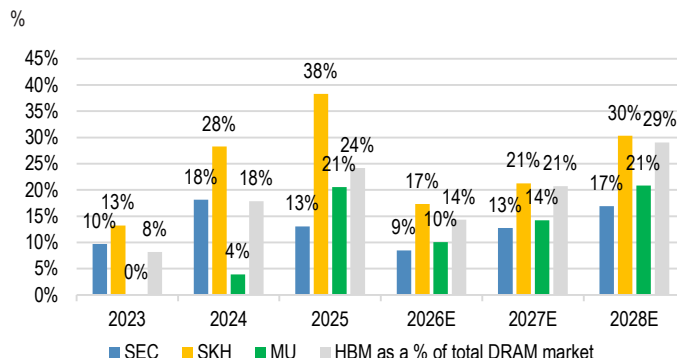
Source: Company data, J.P. Morgan estimates.

Figure 36: HBM market share by sales



Source: Company data, J.P. Morgan estimates.

Figure 37: HBM sales mix by major players



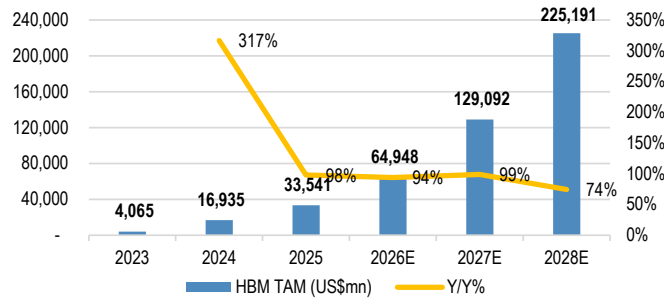
Source: Company data, J.P. Morgan estimates.

HBM demand update

Key changes to GPU shipments include upward revisions to the Blackwell GPU shipments, slightly offset by a slower ramp of the Rubin GPU, leading NVIDIA's units to be revised up to 8.9M/9.9M in 26/27E (vs previously 7.5M/7.6M). Likewise, ASIC unit assumptions include revising up GOOG TPU unit assumptions to 4.5M/8M in 26/27E (vs previously 3.7M/5M) and AMZN Trainium volume to 1.9M/3.3M in 26/27E (vs previously 2.1M/2.6M), with ASIC volumes surpassing merchant GPU unit production by 27E. We expect TPUv8/Trainium 3 to adopt HBM3E given the limited HBM4 supply with the next generation projects such as TPUv9 and Trainium 4 adopting 288GB HBM4. As the majority of the ASIC volumes adopt HBM3E until 27E, we believe this would further squeeze HBM3E S-D creating a favorable market environment for memory makers to raise HBM3E prices. The magnitude of HBM3E price hikes will likely differ across memory makers given uneven contract terms, but we expect HBM3E prices to remain resilient throughout 27E vs previously falling in 26E.

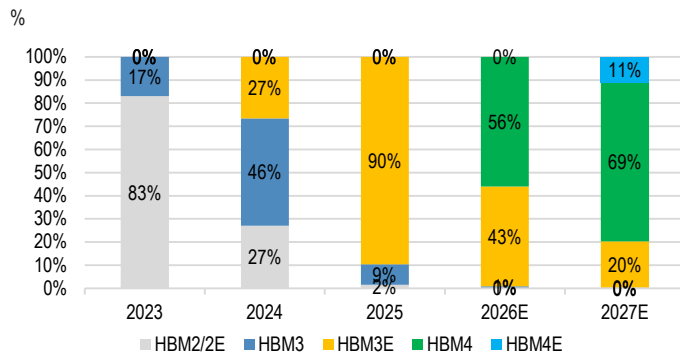
Figure 38: HBM TAM and Y/Y% growth

US\$M, % - RHS



Source: Company data, J.P. Morgan estimates.

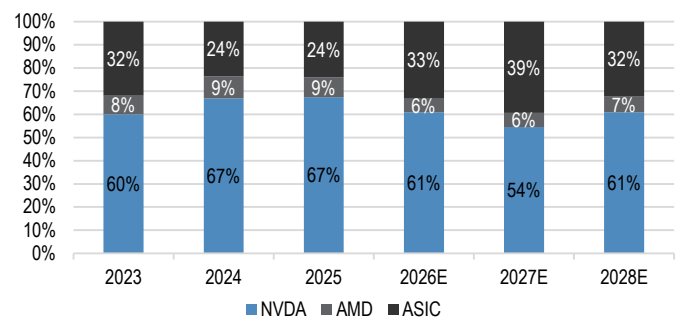
Figure 40: SEC - HBM sales mix by product application



Source: Company data, J.P. Morgan estimates.

Figure 39: HBM bit demand breakdown by customers

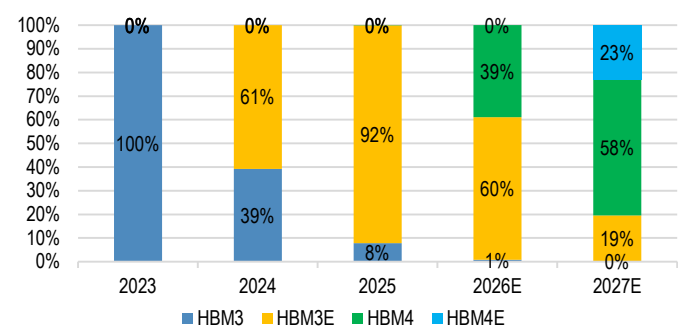
%



Source: Company data, J.P. Morgan estimates.

Figure 41: SKH - HBM sales mix by product application

%

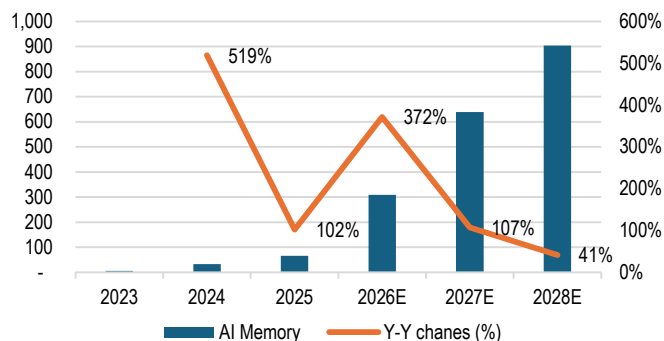


Source: Company data, J.P. Morgan estimates.

AI in Memory

Figure 42: AI memory Y/Y trend

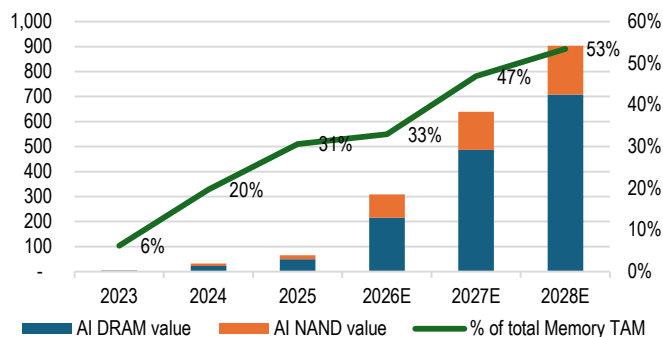
US\$bn, % - RHS



Source: Company reports and J.P. Morgan estimates.

Figure 43: AI DRAM and NAND revenue trend

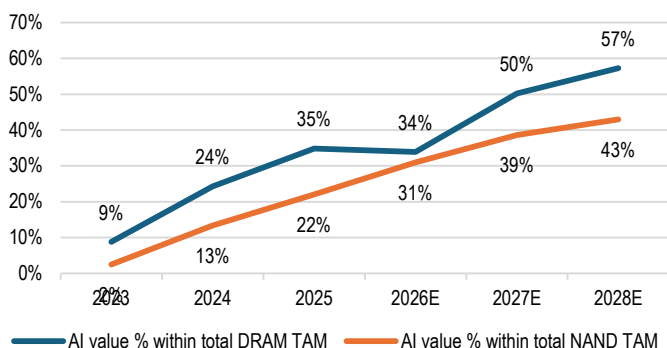
US\$bn, % -RHS



Source: Company reports and J.P. Morgan estimates.

Figure 44: AI value within total DRAM & NAND TAM

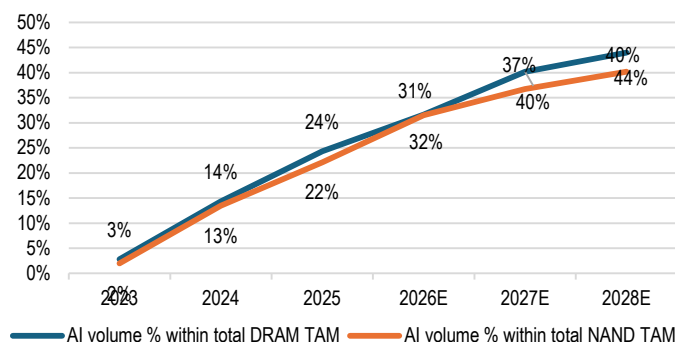
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Source: Company reports and J.P. Morgan estimates.

Figure 45: AI volume with total DRAM & NAND TAM

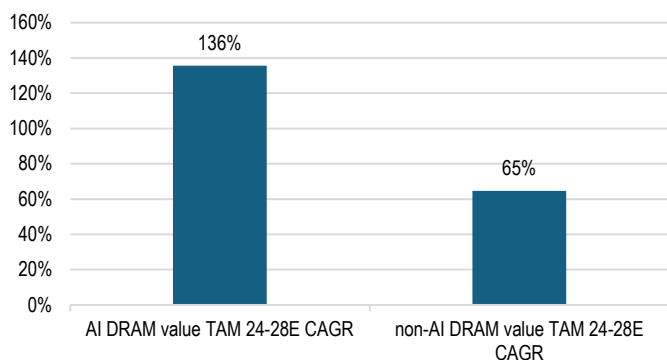
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Source: Company reports and J.P. Morgan estimates.

Figure 46: AI and non-AI DRAM value TAM CAGR trend

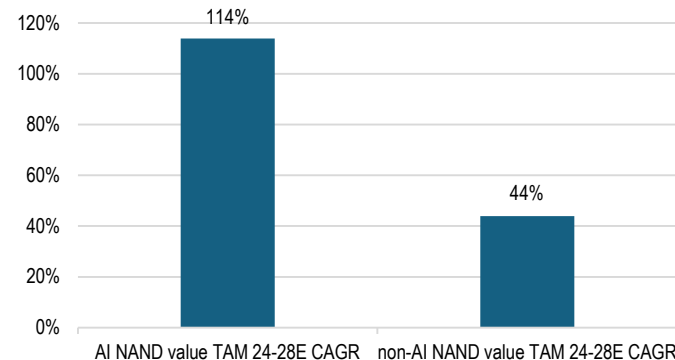
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Source: Company reports and J.P. Morgan estimates.

Figure 47: AI and non-AI NAND value TAM CAGR trend

%



Source: Company reports and J.P. Morgan estimates.

Global memory model: Revisions

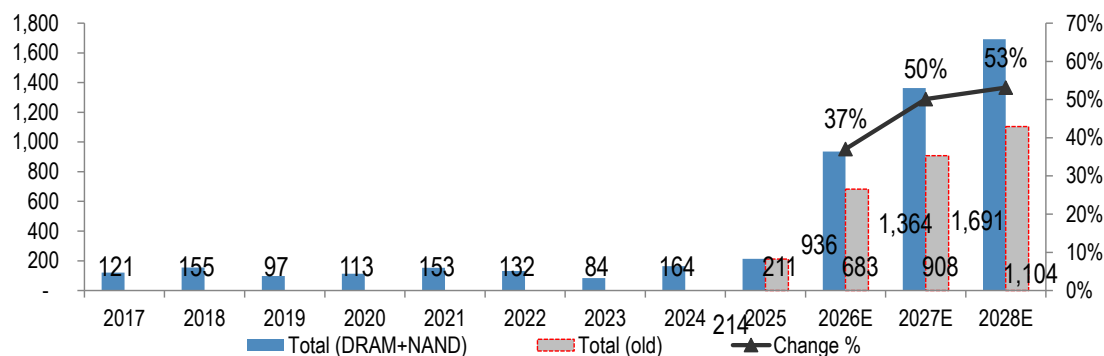
Table 6: J.P. Morgan Global Memory Model: Summary of changes

	Revised			Previous			Difference		
	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E
Memory revenue	936.3	1,363.6	1,691.3	683.2	908.3	1,104.4	37%	50%	53%
DRAM revenue	636.0	970.0	1,236.8	453.7	622.9	774.4	40%	56%	60%
HBM revenue	64.9	129.1	225.2	57.8	98.6	167.2	12%	31%	35%
NAND revenue	300.3	393.6	454.5	229.4	285.5	330.0	31%	38%	38%
Memory price									
DRAM ASP (8Gbe.)	12.9	16.0	16.7	9.4	11.0	11.4	37%	45%	47%
HBM ASP (8Gbe.)	15.5	20.5	23.8	15.3	17.2	19.3	1%	19%	23%
NAND ASP (256Gbe.)	7.4	8.7	8.4	5.8	6.4	6.2	27%	35%	36%
Memory Consumption									
DRAM consumption (8Gbe.)	49,319	60,566	73,927	48,087	56,526	68,135	3%	7%	9%
NAND consumption (256Gbe.)	40,465	45,287	54,279	39,235	44,349	53,554	3%	2%	1%
% growth assumption									
DRAM supply B/G	23%	22%	24%	20%	21%	22%	3%	1%	2%
NAND supply B/G	20%	21%	21%	18%	22%	22%	1%	-1%	-1%
DRAM ASP Y/Y%	246%	24%	4%	161%	17%	3%	86%	7%	1%
NAND ASP Y/Y%	217%	17%	-4%	147%	10%	-4%	70%	7%	1%
Shipments									
Desktops	57	57	58	59	58	62	-3%	-2%	-6%
Notebooks	157	159	162	163	163	173	-3%	-2%	-6%
Servers	15.6	18.9	21.0	14.7	16.3	17.6	6%	16%	20%
Smartphones	1,097	1,064	1,054	1,104	1,071	1,060	-1%	-1%	-1%
SSDs	382	415	446	379	400	428	1%	4%	4%
Average DRAM memory (MB)									
Desktops	10,904	10,451	10,124	11,006	10,866	10,841	-1%	-4%	-7%
Notebooks	13,377	12,918	12,943	13,526	13,660	13,855	-1%	-5%	-7%
Servers	1,796,955	2,428,551	2,872,462	1,801,384	2,346,712	2,826,061	0%	3%	2%
Smartphones	8,864	9,031	9,367	8,687	9,036	9,397	2%	0%	0%
Average NAND memory (MB)									
SSDs	2,035,835	2,711,756	3,336,903	1,983,063	2,558,109	3,072,747	3%	6%	9%
Smartphones	252,713	255,447	257,513	249,442	259,786	279,252	1%	-2%	-8%
Supply assumptions									
DRAM (8Gbe., M)	47,138	57,379	71,035	46,129	55,793	68,174	2%	3%	4%
HBM (8Gbe., M)	4,200	6,311	9,444	4,307	6,383	9,619	-2%	-1%	-2%
NAND (256Gbe., M)	37,106	44,984	54,407	36,208	44,283	54,005	2%	2%	1%
Memory capex	114,882	151,747	183,297	105,429	143,798	169,653	9%	6%	8%
DRAM	91,389	124,029	148,744	84,437	120,673	139,833	8%	3%	6%
NAND	23,493	27,718	34,553	20,992	23,125	29,820	12%	20%	16%

Source: J.P. Morgan estimates.

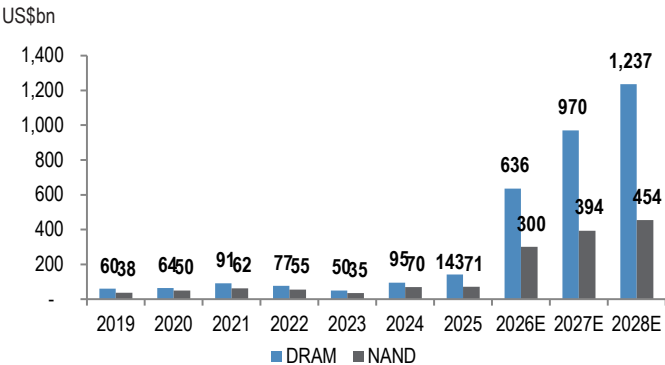
Figure 48: Global memory market (DRAM+NAND) size and % change

US\$bn, % - RHS



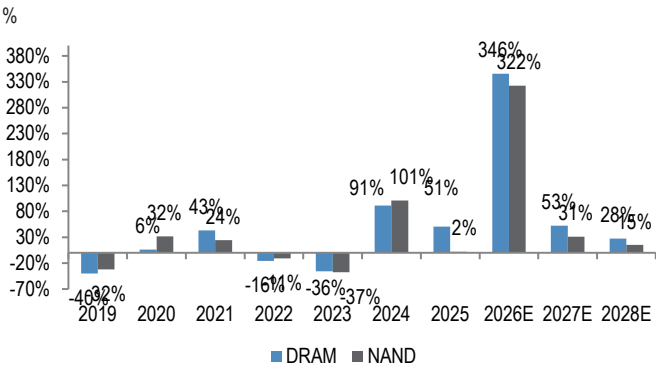
Source: WSTS, J.P. Morgan estimates.

Figure 49: DRAM and NAND market size



Source: WSTS, J.P. Morgan estimates.

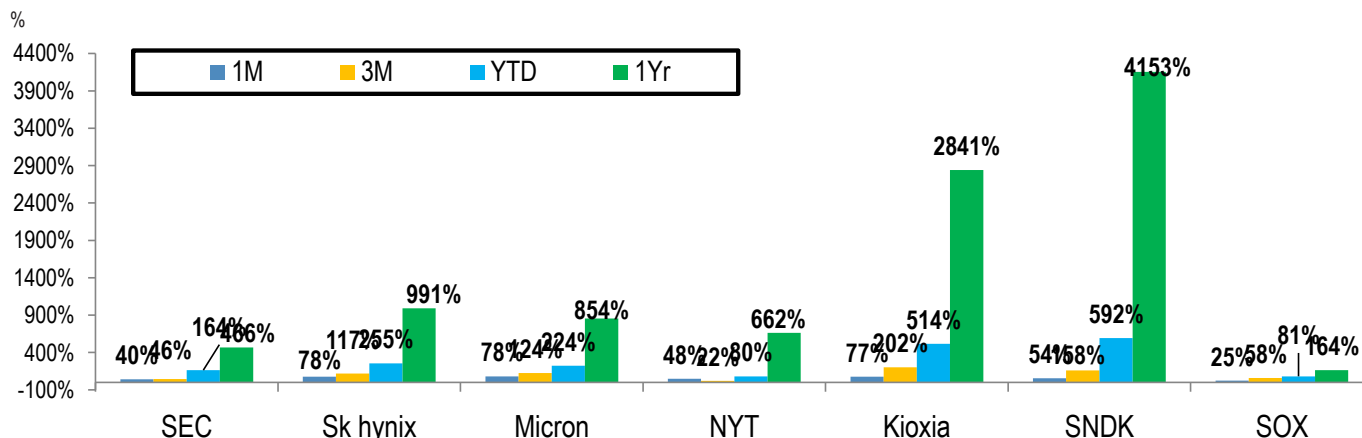
Figure 50: DRAM and NAND market growth rates



Source: WSTS, J.P. Morgan estimates.

Share price related charts

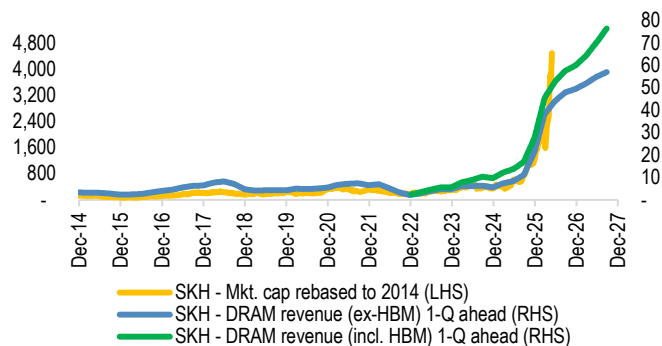
Figure 51: Global memory makers' share price performance including SOX (Philadelphia Semiconductor index)



Source: Bloomberg Finance L.P. Note: Past performance is not an indicator of future results

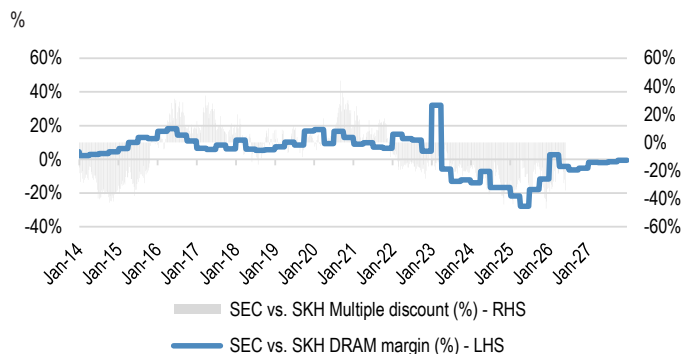
Figure 52: SKH market cap vs DRAM revenue comparison (1Q ahead)

Rebased as of 1/1/2014, US\$bn - RHS



Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates. Note: 2Q26 onwards are J.P. Morgan estimates.

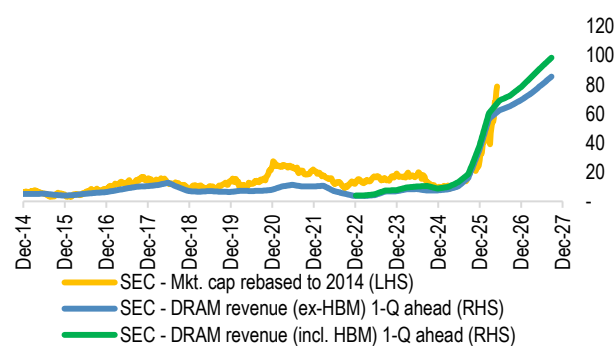
Figure 54: SEC vs SKH: DRAM margin difference vs FTM P/B multiple discount rate



Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates. Note: 2Q26 onwards are J.P. Morgan estimates.

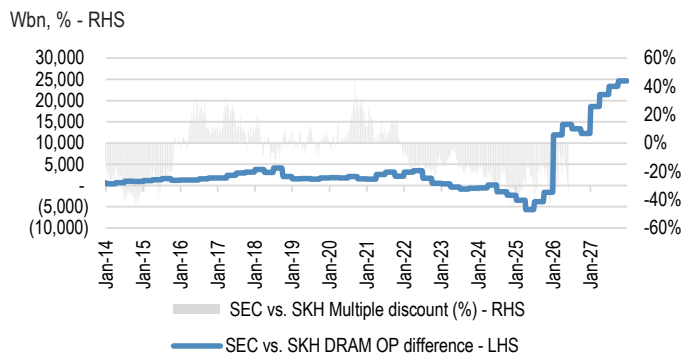
Figure 53: SEC market cap vs DRAM revenue comparison (1Q ahead)

Rebased as of 1/1/2014, US\$bn - RHS



Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates. Note: 2Q26 onwards are J.P. Morgan estimates.

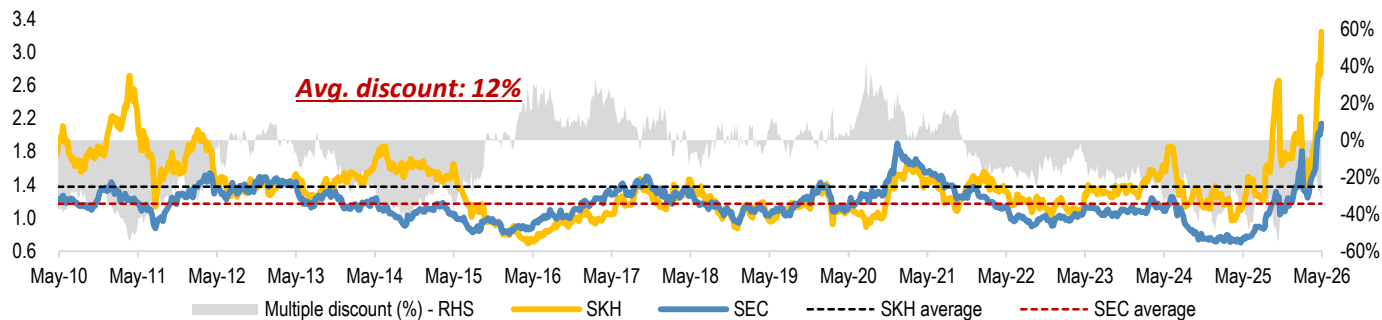
Figure 55: SEC vs SKH: absolute DRAM OP difference vs FTM P/B multiple discount rate



Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates. Note: 2Q26 onwards are J.P. Morgan estimates.

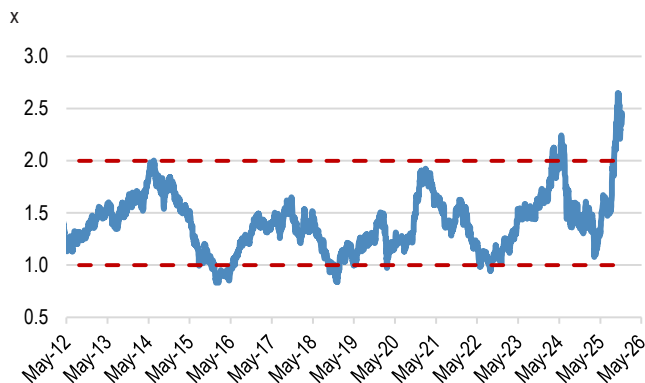
Figure 56: 12M FWD P/B multiple comparison between SKH and SEC

x, % - RHS



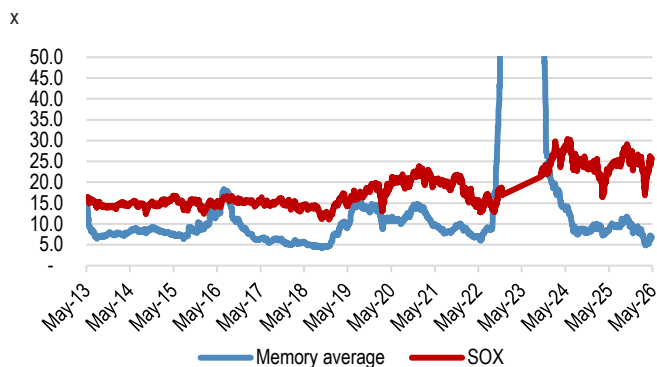
Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates.

Figure 57: Average P/B multiple for top 3 DRAM makers



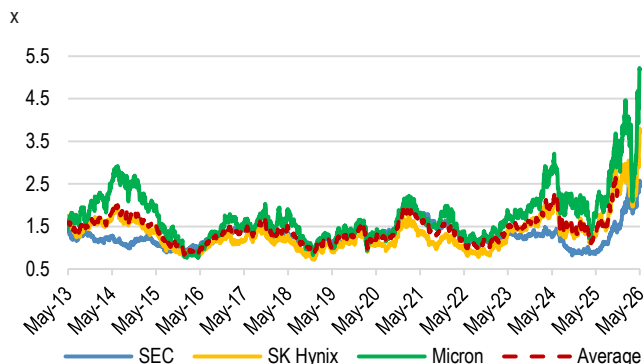
Source: Bloomberg Finance L.P.

Figure 59: Average P/E multiple for top 3 DRAM makers vs. SOX



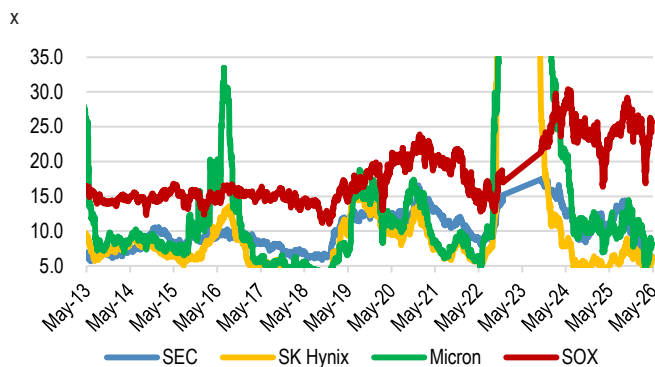
Source: Bloomberg Finance L.P. Note: 23/24 not meaningful from loss making.

Figure 58: 12M Forward P/B multiple trend for DRAM makers



Source: Bloomberg Finance L.P.

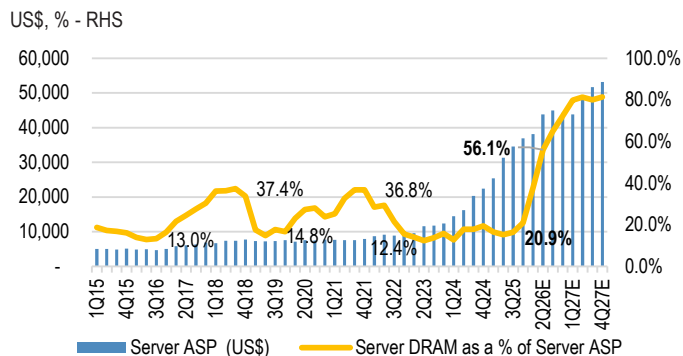
Figure 60: 12M Forward P/E multiple trend for DRAM makers vs. SOX



Source: Bloomberg Finance L.P. Note: 23/24 not meaningful from loss making.

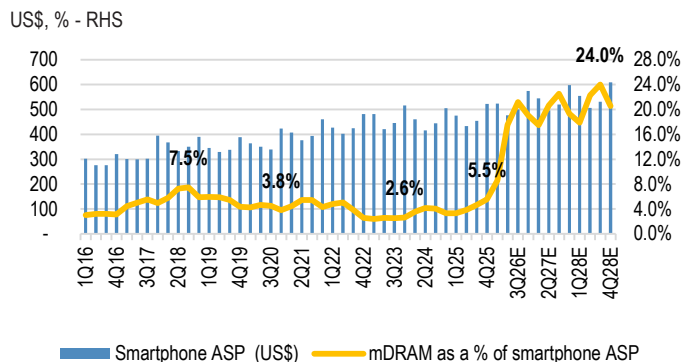
Memory TCO update on major end applications

Figure 61: Server DRAM cost as a % of server ASP



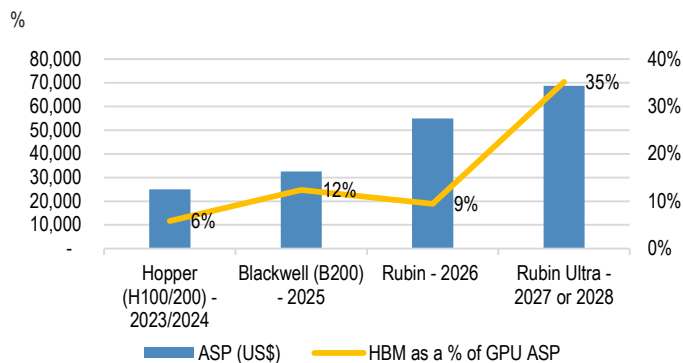
Source: IDC, J.P. Morgan estimates.

Figure 63: mDRAM cost as a % of smartphone ASP



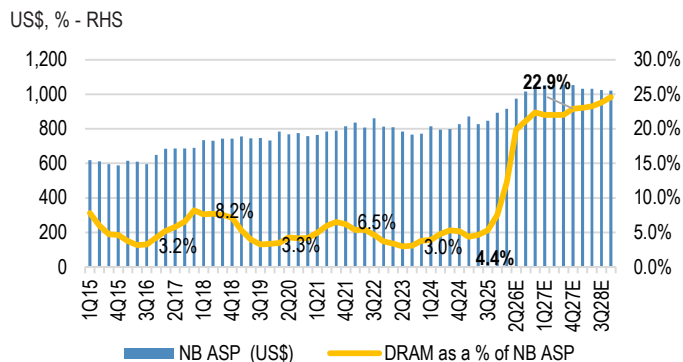
Source: OMDIA, J.P. Morgan estimates.

Figure 62: HBM as a % of GPU chip ASP



Source: Company data, J.P. Morgan estimates.

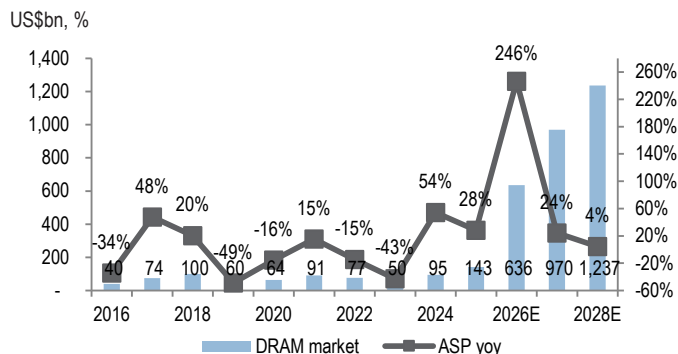
Figure 64: DRAM cost as a % of NB ASP



Source: IDC, J.P. Morgan estimates.

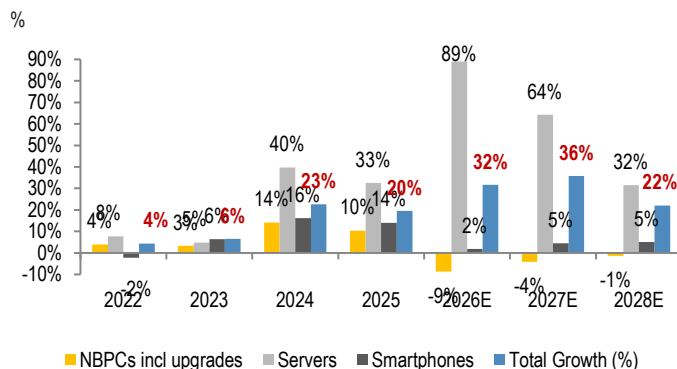
DRAM update

Figure 65: DRAM market revenue and ASP YoY change



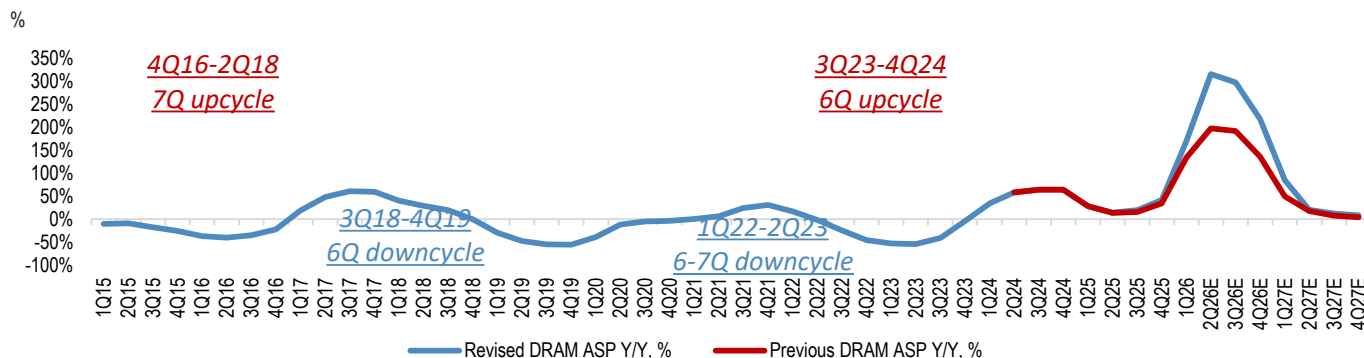
Source: WSTS, J.P. Morgan estimates

Figure 66: DRAM bit demand growth Y/Y by key applications



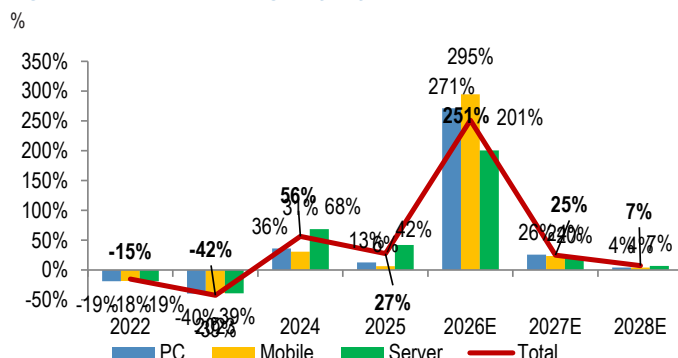
Source: Gartner®, IDC, OMDIA, J.P. Morgan estimates. Note: Server includes HBM.

Figure 67: DRAM ASP Y/Y% change vs. previous assumptions



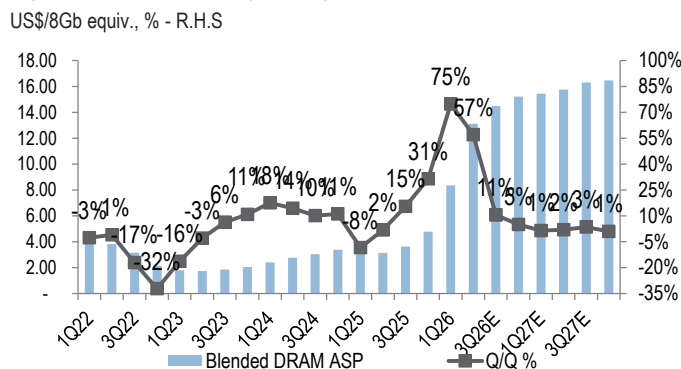
Source: WSTS, J.P. Morgan estimates.

Figure 68: DRAM ASP changes by key applications



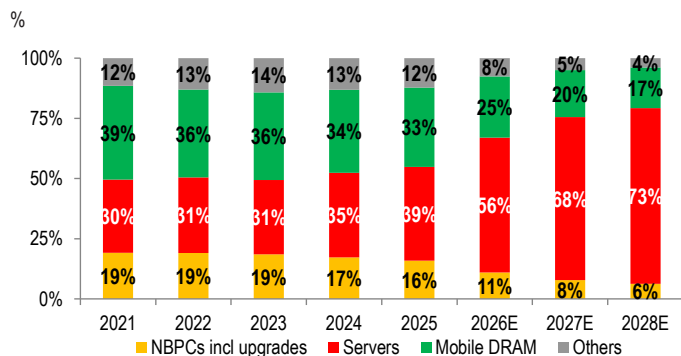
Source: OMDIA, WSTS, J.P. Morgan estimates.

Figure 69: DRAM quarterly pricing trend and Q/Q trend



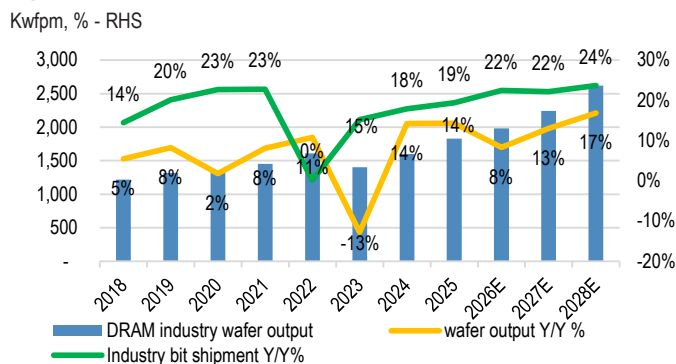
Source: OMDIA, WSTS, J.P. Morgan estimates.

Figure 70: DRAM bit demand mix



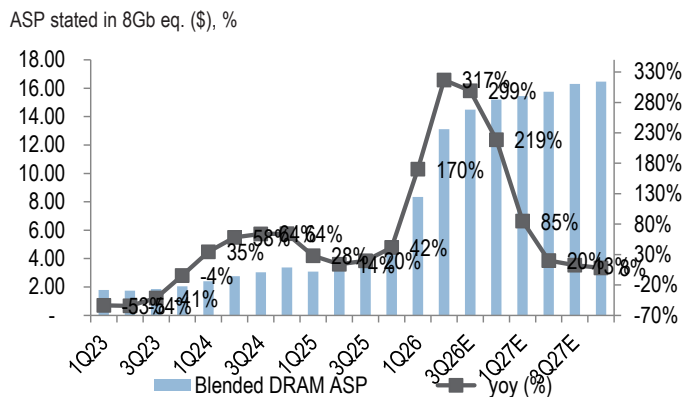
Source: Gartner®, J.P. Morgan estimates.

Figure 72: DRAM WSPM and bit shipment Y/Y% trend



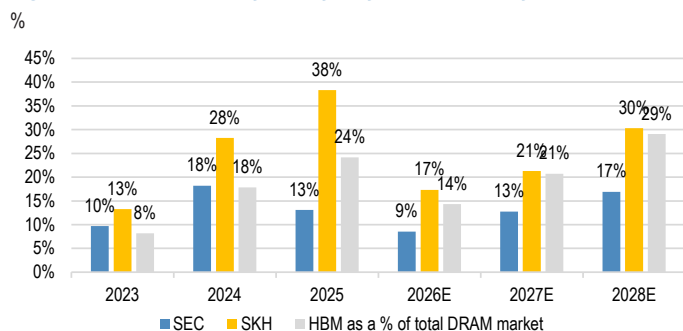
Source: Company data, OMDIA, J.P. Morgan estimates.

Figure 71: Blended quarterly DRAM ASP and Y/Y trend



Source: WSTS, J.P. Morgan estimates.

Figure 73: HBM % capacity mix by players and industry

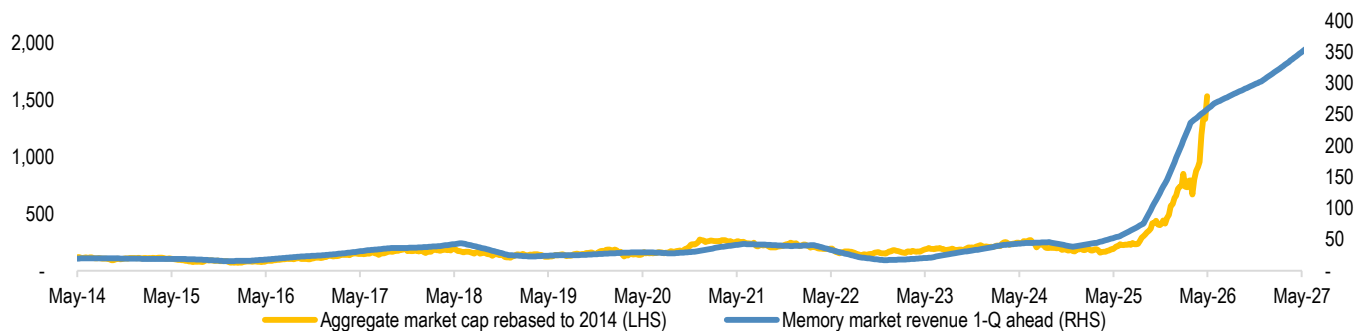


Source: Company data, J.P. Morgan estimates.

Memory market cap vs key financial indicators

Figure 74: Aggregate market cap vs leading memory market revenue trend (1Q ahead)

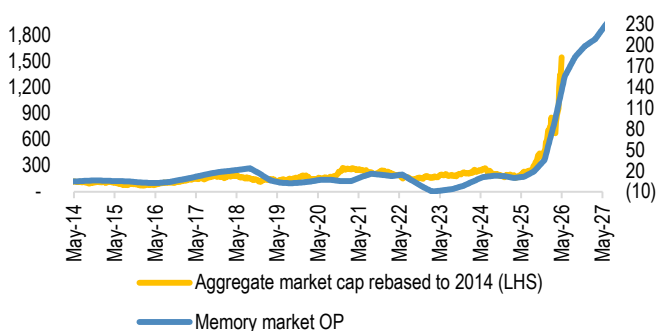
Rebased as of 1/1/2014, US\$bn-RHS



Source: WSTS, Bloomberg Finance L.P., J.P. Morgan estimates. Note: Market cap includes SEC, SK Hynix, Micron, Nanya Technology, Western Digital. Kioxia / SNDK added from 1Q25 and excluded WDC. Note: 2Q26 onwards are J.P. Morgan estimates.

Figure 75: Aggregate market cap vs. memory industry absolute OP

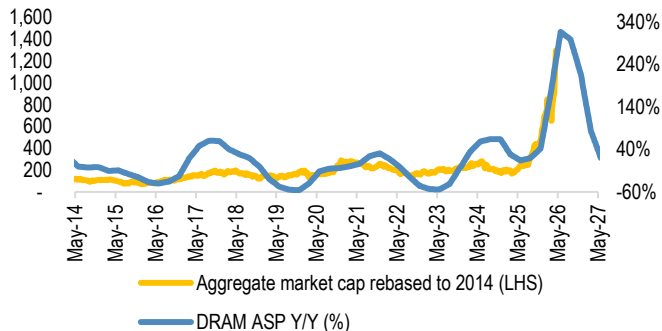
Rebased as of 1/1/2014, US\$bn-RHS



Source: WSTS, OMDIA, Bloomberg Finance L.P., J.P. Morgan estimates. Note: Market cap includes SEC, SK Hynix, Micron, Nanya Technology, Western Digital. Kioxia / SNDK added from 1Q25 and excluded WDC. Note: 2Q26 onwards are J.P. Morgan estimates.

Figure 76: Aggregate market cap vs. DRAM ASP Y/Y change

Rebased as of 1/1/2014, % - RHS

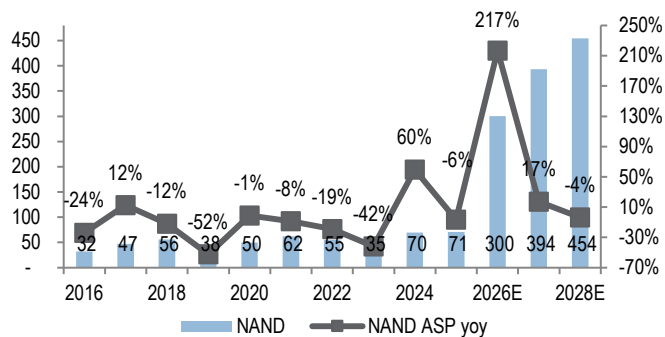


Source: WSTS, Bloomberg Finance L.P., J.P. Morgan estimates. Note: Market cap includes SEC, SK Hynix, Micron, Nanya Technology. Note: 2Q26 onwards are J.P. Morgan estimates.

NAND update

Figure 77: NAND market revenue and ASP YoY change

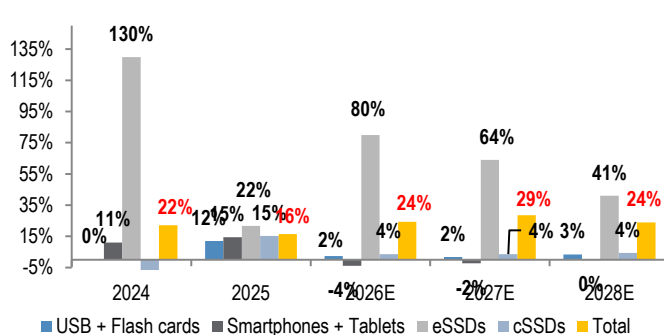
US\$bn, %



Source: WSTS, J.P. Morgan estimates.

Figure 78: NAND demand growth Y/Y by key application

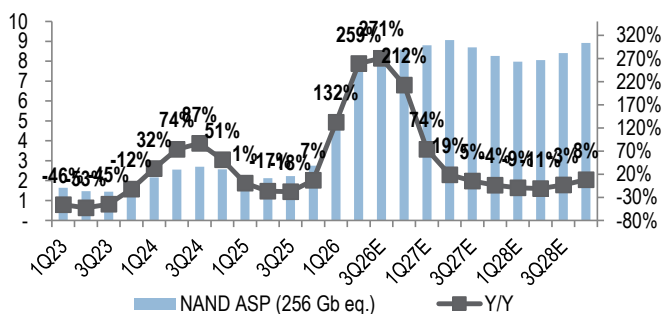
%



Source: OMDIA, WSTS, J.P. Morgan estimates.

Figure 79: NAND quarterly pricing trend and Y/Y trend

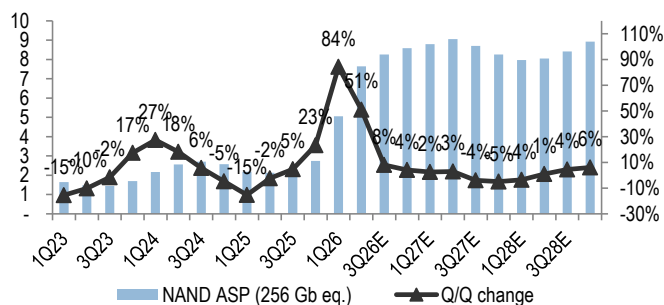
ASP stated in 256Gb eq. (\$), %



Source: WSTS, J.P. Morgan estimates.

Figure 80: NAND quarterly pricing trend and Q/Q trend

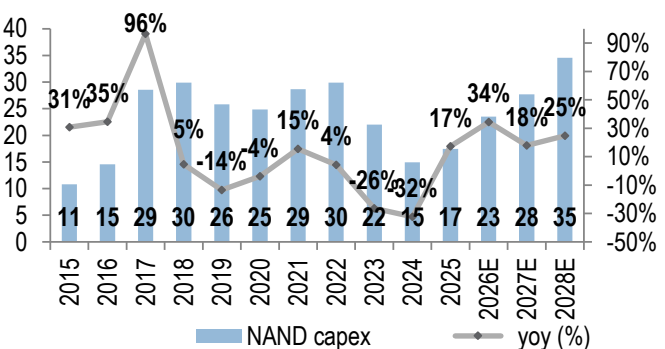
ASP stated in 256Gb eq. (\$), %



Source: WSTS, J.P. Morgan estimates.

Figure 81: NAND capex and Y/Y trend

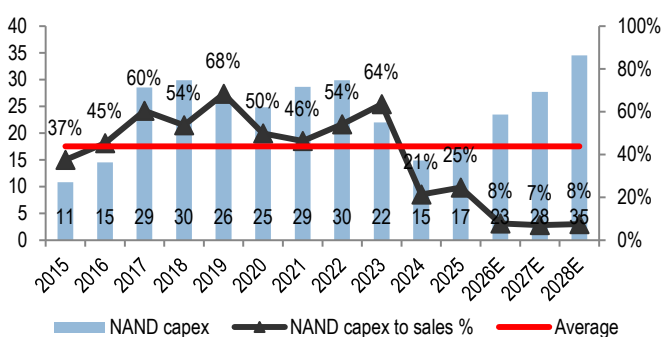
US\$bn, %



Source: Company data, J.P. Morgan estimates.

Figure 82: NAND capital intensity

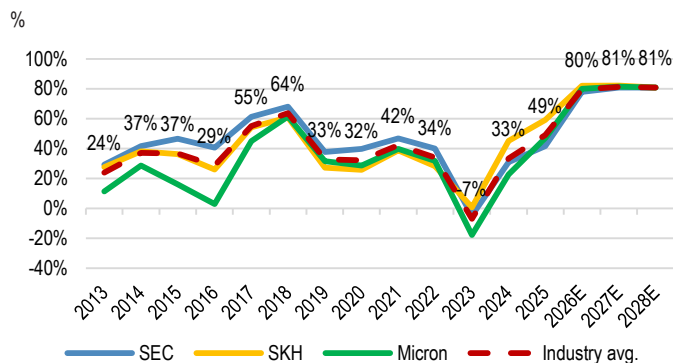
US\$bn, %



Source: Company data, J.P. Morgan estimates. Note: Average from 2011-23.

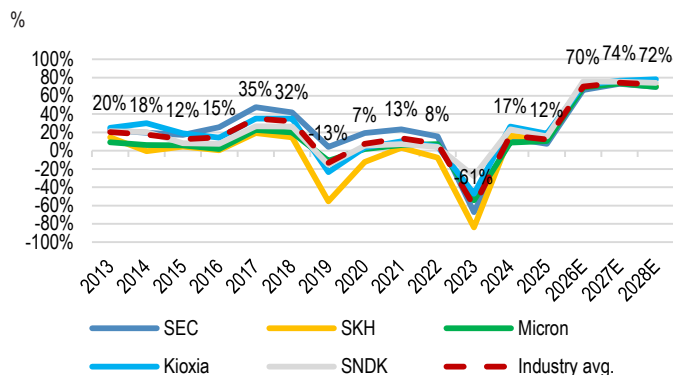
Memory OPM and spot price trend

Figure 83: DRAM OPM trend



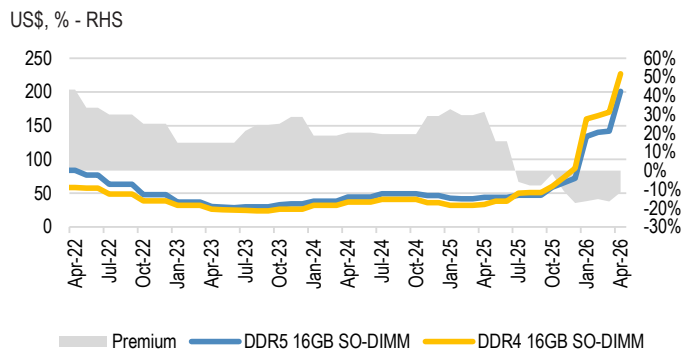
Source: OMDIA, company data, J.P. Morgan estimates.

Figure 84: NAND OPM trend



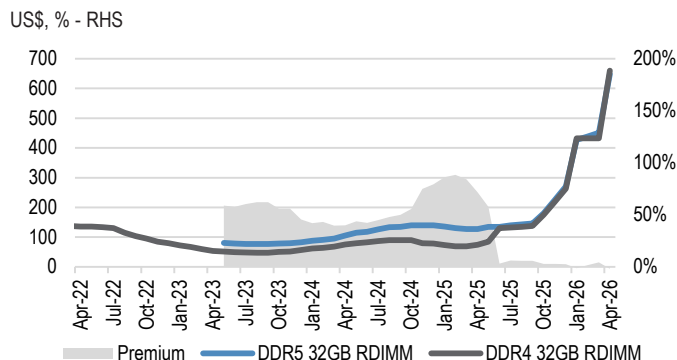
Source: OMDIA, company data, J.P. Morgan estimates.

Figure 85: DDR5 vs DDR4 SO-DIMM price trend



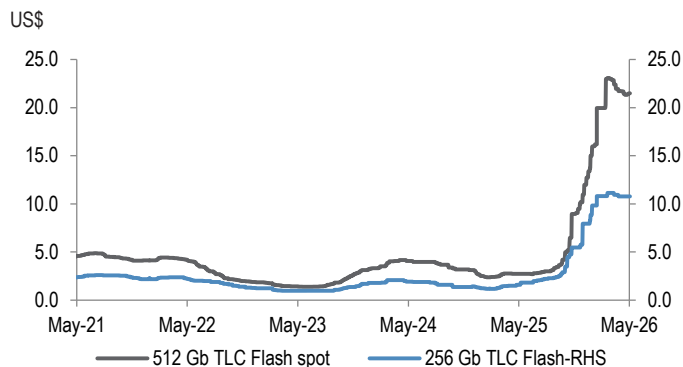
Source: Company data.

Figure 86: DDR5 vs DDR4 RDIMM price trend



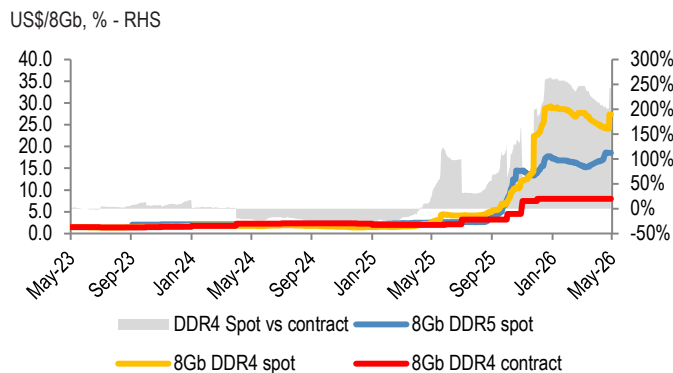
Source: Company data.

Figure 87: NAND spot price trend



Source: Bloomberg Finance L.P.

Figure 88: 8Gb DDR4 spot vs contract price trend



Source: Bloomberg Finance L.P.

Appendix

Global DRAM model

Table 7: Global DRAM supply-demand model

8Gb. equivalent

	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Total demand (M units)	8,266	9,031	10,045	10,577	10,925	12,026	13,254	14,021	15,304	16,461	17,511	18,227	31,364	37,920	50,226	67,504	81,771
Sequential growth (%)	-3%	9%	11%	5%	3%	10%	10%	6%	9%	8%	6%	4%	22%	21%	32.5%	34.4%	21.1%
Y/Y growth (%)	17%	19%	22%	25%	32%	33%	32%	33%	40%	37%	32%	30%			9.5%	12.7%	-2.7%
Total supply (M units)	8,024	9,408	10,349	10,569	11,026	11,653	12,152	12,307	12,888	13,736	14,917	15,838	32,131	38,351	47,138	57,379	71,035
Sequential growth (%)	-1%	17%	10%	2%	4%	6%	4%	1%	5%	7%	9%	6%	18%	19%	22.9%	21.7%	23.8%
Y/Y growth (%)	5%	15%	27%	30%	37%	24%	17%	16%	17%	18%	23%	29%					
Consumption (WSTS)	8,019	9,254	10,286	10,781	11,581	12,165	12,640	12,933	13,708	14,608	15,728	16,522	32,669	38,340	49,319	60,566	73,927
Sequential growth (%)	-3%	15%	11%	5%	7%	5%	4%	2%	6%	7%	8%	5%	24%	17%	28.6%	22.8%	22.1%
Y/Y growth (%)	3%	11%	24%	30%	44%	31%	23%	20%	18%	20%	24%	28%					
Inventory build-up	6	154	63	(212)	(555)	(512)	(487)	(627)	(821)	(872)	(811)	(684)	-	11	-	-	-
Cumulative inventory	1,656	1,810	1,873	1,661	1,106	594	107	(520)	(1,341)	(2,213)	(3,023)	(3,707)	1,650	1,661	(520)	(3,707)	(6,599)
Sufficiency ex inventory (%)	0%	2%	1%	-2%	-5%	-4%	-4%	-5%	-6%	-6%	-5%	-4%	-1.6%	0.0%	-4.4%	-5.3%	-3.9%
Sufficiency (%)	5.2%	5.3%	4.7%	3.4%	1.2%	0.2%	-0.8%	-2.2%	-3.9%	-5.3%	-6.1%	-6.6%	3.4%	4.4%	-5.5%	-11.4%	-12.8%
Inventory weeks	2.6	2.8	2.7	2.3	1.4	0.7	0.1	(0.6)	(1.4)	(2.3)	(2.9)	(3.4)					
DRAM Revenues (US\$, M)	24,773	29,104	37,352	51,478	96,707	159,479	183,101	196,727	211,643	230,039	256,346	271,976	94,764	142,707	636,015	970,004	1,236,783
Sequential growth (%)	-11%	17%	28%	38%	88%	65%	15%	7%	8%	9%	11%	6%	91%	51%	346%	53%	28%
Y/Y growth (%)	32%	27%	49%	85%	290%	448%	390%	282%	119%	44%	40%	38%					
DRAM ASP (\$)	3.09	3.15	3.63	4.77	8.35	13.11	14.49	15.21	15.44	15.75	16.30	16.46	2.90	3.72	12.90	16.02	16.73
Sequential growth (%)	-8%	2%	15%	31%	75%	57.0%	10.5%	5.0%	1.5%	2.0%	3.5%	1.0%	54%	28%	246%	24%	4%
Y/Y growth (%)	28%	14%	20%	42%	170%	317%	299%	219%	85%	20%	13%	8%					
	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Consumption vs. Demand	3%	-2%	-2%	-2%	-6%	-1%	5%	8%	12%	13%	11%	10%	-4%	-1%	2%	11%	11%
Consumption vs. Supply	0%	2%	1%	-2%	-5%	-4%	-4%	-5%	-6%	-6%	-5%	-4%	-2%	0%	-4%	-5%	-4%

Source: WSTS, J.P. Morgan estimates, Gartner®.

Table 8: DRAM demand model

8Gb. equivalent (millions)

Demand (8Gb. Eq., M)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
PC DRAM	1,312	1,400	1,569	1,602	1,441	1,332	1,318	1,282	1,231	1,249	1,302	1,369	5,332	5,883	5,374	5,151	5,081
Desktop	151	169	191	206	160	151	141	155	125	137	142	180	617	718	606	584	574
Notebooks	500	551	653	642	558	486	511	500	467	472	521	549	2,050	2,346	2,055	2,009	2,051
Upgrades & other data processing	660	680	725	754	724	695	667	627	640	640	640	640	2,665	2,819	2,713	2,558	2,456
Server	3,001	3,296	3,775	4,382	5,408	6,342	7,546	8,014	9,871	10,931	11,736	12,315	10,902	14,454	27,309	44,853	58,985
AI server (incl. SOCAMM)	343	402	463	505	757	911	1,655	1,783	2,928	3,100	3,410	3,675	852	1,713	5,105	13,113	18,140
General purpose server	2,266	2,413	2,725	3,208	3,923	4,570	4,851	5,163	5,726	6,506	6,829	6,999	8,806	10,612	18,507	26,060	32,346
HBM	392	481	587	669	728	861	1,039	1,067	1,217	1,326	1,496	1,641	1,244	2,129	3,696	5,680	8,500
Mobile DRAM	2,768	2,830	3,278	3,326	2,708	2,628	2,995	2,967	2,694	2,629	2,952	2,919	10,683	12,201	11,298	11,193	11,500
Others	995	1,123	1,174	1,234	1,085	1,188	1,239	1,303	1,136	1,241	1,295	1,361	4,070	4,526	4,815	5,033	5,269
Consumer Electronics	429	506	514	535	446	516	524	546	451	521	529	551	1,830	1,984	2,032	2,052	2,073
Automotive Electronics	195	208	225	231	244	237	252	259	273	266	282	290	675	859	992	1,111	1,244
Industrial Electronics	310	332	353	375	329	352	374	397	342	366	389	413	1,280	1,370	1,452	1,510	1,570
Military/Civil Aerospace	61	77	83	94	67	83	89	101	71	88	94	107	285	314	340	360	382
Total	8,075	8,648	9,796	10,545	10,642	11,490	13,098	13,566	14,932	16,050	17,285	17,964	30,987	37,064	48,796	66,230	80,836

Source: Gartner®, J.P. Morgan estimates

Table 9: DRAM supply model

8Gb. equivalent (millions)

DRAM Supply (8Gb. Eq., M)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
SK hynix	2,242	2,795	3,013	3,028	3,014	3,249	3,427	3,534	3,642	3,790	4,065	4,356	8,933	11,077	13,224	15,853	19,024
Samsung	3,200	3,552	4,102	4,185	4,308	4,532	4,672	4,665	4,937	5,248	5,672	6,112	13,890	15,039	18,177	21,969	26,988
Micron	1,950	2,313	2,369	2,400	2,538	2,646	2,730	2,793	2,823	3,079	3,356	3,387	7,375	9,031	10,707	12,646	15,384
Nanya	92	152	192	216	203	191	184	180	180	189	199	226	421	652	758	795	1,012
Powerchip	44	50	54	54	53	60	64	64	64	72	77	77	167	202	242	290	348
Others	450	500	570	630	840	892	961	944	1,134	1,234	1,414	1,547	1,215	2,150	3,637	5,329	7,692
Total	8,024	9,408	10,349	10,569	11,026	11,653	12,152	12,307	12,888	13,736	14,917	15,838	32,131	38,351	46,968	57,379	70,965

Source: OMDIA, company data, J.P. Morgan estimates

Table 10: DRAM capex by company

US\$ mn

	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
SK hynix	3,600	3,851	6,818	7,162	5,204	6,522	8,705	4,294	10,034	17,314	26,409	37,493	45,986
Samsung	3,882	6,018	11,819	7,291	9,204	14,008	13,652	16,702	19,083	19,824	30,423	42,143	51,071
Micron	2,100	2,800	4,400	4,000	3,250	5,000	5,626	3,300	5,500	13,300	23,200	29,800	34,700
Nanya	693	961	669	178	286	400	695	428	505	431	1,644	1,644	1,138
Winbond	79	250	273	304	267	349	1,415	445	537	208	632	632	632
Powerchip	40	50	357	250	188	200	230	180	180	150	180	216	216
Others (incl. China)	-	-	500	550	1,000	2,000	3,000	5,500	7,000	8,000	8,900	12,100	15,000
Total	11,294	13,930	24,836	19,735	19,399	28,479	33,323	30,849	42,839	59,226	91,389	124,029	148,744
Change (%)	-20%	23%	78%	-21%	-2%	47%	17%	-7%	39%	38%	54%	36%	20%

Source: OMDIA, company data, J.P. Morgan estimates

Global NAND model

Table 11: Global NAND supply-demand model

256 Gb. equivalent

256Gb. Eq. (units in millions)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Demand	6,342	7,189	8,274	8,905	8,744	9,156	9,723	10,165	10,808	11,667	12,780	13,661	25,746	30,710	37,788	48,916	59,817
Sequential change (%)	-7%	13%	15%	8%	-2%	5%	6%	5%	6%	8%	10%	7%	20%	19%	23.0%	29.4%	22.3%
Y/Y change (%)	9%	16%	20%	30%	38%	27%	18%	14%	24%	27%	31%	34%			3.3%	8.2%	1.3%
Supply	6,216	7,670	8,458	8,639	8,581	8,995	9,561	9,970	10,148	10,837	11,737	12,261	27,060	30,982	37,106	44,984	54,407
Sequential change (%)	-8%	23%	10%	2%	-1%	5%	6%	4%	2%	7%	8%	4%	14%	14%	19.8%	21.2%	20.9%
Y/Y change (%)	-7%	14%	24%	27%	38%	17%	13%	15%	18%	20%	23%	23%					
Consumption	6,254	7,317	8,120	8,669	9,683	10,119	10,269	10,394	10,399	10,999	11,737	12,151	27,836	30,360	40,465	45,287	54,279
Sequential change (%)	-10%	17%	11%	7%	12%	4%	1%	1%	0%	6%	7%	4%	26%	9.1%	33.3%	11.9%	19.9%
Y/Y change (%)	-9%	4%	15%	25%	55%	38%	26%	20%	7%	9%	14%	17%					
Inventory build-up	(38)	352	338	(30)	(1,103)	(1,124)	(708)	(425)	(252)	(161)	(0)	110	-	622	-	-	128
Cumulative inventory build up	3,757	4,109	4,447	4,417	3,315	2,191	1,483	1,058	807	645	645	755	3,795	4,417	1,058	755	883
Sufficiency (ex inventory)	-1%	5%	4%	0%	-11%	-11%	-7%	-4%	-2%	-1%	0%	1%	-3%	2%	-8%	-1%	0%
Sufficiency (incl. inventory)	14.9%	15.2%	14.7%	12.7%	5.7%	2.6%	1.9%	1.5%	1.3%	1.1%	1.4%	1.8%	10.8%	16.6%	-5.7%	1.0%	1.9%
Inventory weeks	7.4	7.8	7.9	7.4	5.2	3.3	2.2	1.5	1.1	0.8	0.8	0.9					
		-0.4%	0.5%	2.1%	6.9%	3.1%	0.7%	0.4%	0.2%	0.2%	-0.3%	-0.4%					
NAND revenues (US\$, M)	13,626	15,558	18,077	23,815	49,000	77,319	84,739	89,208	91,482	99,657	102,096	100,410	69,691	71,077	300,265	393,645	454,485
Sequential change (%)	-23%	14%	16%	32%	106%	58%	10%	5%	3%	9%	2%	-2%	101%	2%	322%	31%	15%
Y/Y change (%)	-8%	-14%	-5%	34%	260%	397%	369%	275%	87%	29%	20%	13%					
NAND ASP (256Gb. Eq.)	2.18	2.13	2.23	2.75	5.06	7.64	8.25	8.58	8.80	9.06	8.70	8.26	2.50	2.34	7.42	8.69	8.37
Sequential change (%)	-15%	-2%	5%	23%	84%	51.0%	8.0%	4.0%	2.5%	3.0%	-4.0%	-5.0%	60%	-6%	217%	17%	-4%
Y/Y change (%)	1%	-17%	-18%	7%	132%	259%	271%	212%	74%	19%	5%	-4%					
	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Demand vs. Consumption	1%	-2%	2%	3%	-10%	-10%	-5%	-2%	4%	6%	9%	12%	-8%	1%	-7%	8%	10%
Supply vs. Consumption	-1%	5%	4%	0%	-11%	-11%	-7%	-4%	-2%	-1%	0%	1%	-3%	2%	-8%	-1%	0%
Demand vs. Supply	2%	-6%	-2%	3%	2%	2%	2%	2%	7%	8%	9%	11%	-5%	-1%	2%	9%	10%

Source: OMDIA, company data, J.P. Morgan estimates

Table 12: NAND demand model

256 Gb. equivalent (millions)

NAND demand (256Gb. Eq., M)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
USB	151	162	166	161	156	164	166	166	160	166	164	167	564	639	652	657	683
Flash cards	151	156	137	142	157	160	141	145	160	163	147	150	529	586	603	620	639
SSDs	3,096	3,767	4,300	5,172	5,323	5,726	6,268	6,426	7,257	8,114	9,053	9,885	13,769	16,335	23,744	34,309	45,460
Enterprise SSD	1,531	1,990	2,322	3,091	3,527	3,861	4,265	4,429	5,484	6,182	6,888	7,818	7,344	8,934	16,082	26,372	37,174
Client SSD/Others	1,565	1,776	1,978	2,080	1,796	1,865	2,003	1,997	1,773	1,933	2,165	2,067	6,424	7,400	7,661	7,937	8,285
Smartphones	1,881	1,886	2,387	2,606	2,147	1,968	2,096	2,251	2,081	1,887	2,077	2,252	7,607	8,761	8,462	8,297	8,281
Tablets	200	225	245	256	191	212	219	225	177	196	212	219	850	926	848	805	812
Feature phones	2	2	2	2	2	2	2	2	2	2	2	2	8	9	8	8	8
Others	614	639	658	693	674	699	711	749	731	758	771	812	2,310	2,603	2,832	3,072	3,340
Total	6,095	6,836	7,895	9,032	8,651	8,932	9,603	9,963	10,569	11,287	12,426	13,488	25,636	29,858	37,149	47,769	59,222

Source: IDC, Gartner®, J.P. Morgan estimates

Table 13: NAND supply model

256 Gb. equivalent (millions)

NAND supply (256Gb. Eq., M)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
SK hynix	854	1,447	1,388	1,547	1,377	1,569	1,648	1,681	1,748	1,853	1,927	1,966	4,749	5,236	6,275	7,494	8,858
Samsung	1,821	2,304	2,522	2,295	2,502	2,552	2,654	2,813	2,870	2,984	3,193	3,321	9,194	8,942	10,522	12,368	14,678
Micron	767	959	920	985	1,000	1,050	1,103	1,169	1,192	1,240	1,289	1,360	2,974	3,630	4,321	5,081	6,100
Kioxia	1,141	1,144	1,555	1,617	1,453	1,511	1,738	1,807	1,699	1,835	2,073	2,177	4,738	5,456	6,510	7,785	9,285
SNDK	903	956	1,102	1,135	1,169	1,192	1,246	1,277	1,316	1,408	1,506	1,612	3,286	4,097	4,885	5,841	6,994
YMTC	730	860	970	1,060	1,080	1,120	1,172	1,222	1,324	1,518	1,748	1,826	2,120	3,620	4,595	6,415	8,491
Total	6,216	7,670	8,458	8,639	8,581	8,995	9,561	9,970	10,148	10,837	11,737	12,261	27,060	30,982	37,106	44,984	54,407

Source: OMDIA, company data, J.P. Morgan estimates

Table 14: NAND capex by company

US\$ mn

	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
SK hynix	1,400	4,072	7,569	4,632	3,303	4,268	5,733	1,936	1,463	1,548	2,611	3,211	5,418
Samsung	4,744	11,948	6,364	6,605	9,494	10,931	9,677	8,798	5,652	5,483	6,223	7,143	8,214
Micron	3,100	2,700	4,996	3,981	3,500	4,700	5,000	3,000	3,000	3,000	4,300	4,800	7,800
Solidigm	1,600	2,500	2,763	2,951	405	480	-	-	-	-	-	-	-
Kioxia + SanDisk	3,700	4,850	5,200	5,500	5,800	5,800	6,000	5,300	2,300	3,450	5,059	5,565	6,121
YMTC		2,500	3,000	2,180	2,350	2,500	3,500	3,000	2,500	4,000	5,300	7,000	7,000
Total	14,544	28,570	29,891	25,849	24,852	28,679	29,910	22,034	14,914	17,481	23,493	27,718	34,553

Source: OMDIA, Company data, J.P. Morgan estimates.

Companies Discussed in This Report (all prices in this report as of market close on 29 May 2026, unless otherwise indicated) Hanmi Semiconductor(042700.KS/W283,000/UW), KIOXIA Holdings (285A)(285A.T/¥65,850/OW), Micron Technology(MU/\$923.52[28 May 2026]/OW), SK hynix(000660.KS/W2,301,000/OW), Samsung Electronics(005930.KS/W317,500/OW), Silicon Motion(SIMO/\$282.44[28 May 2026]/OW), Tokyo Electron (8035)(8035.T/¥52,420/OW), Winbond(2344.TW/NT\$158.00/OW)

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Hanmi Semiconductor (042700.KS, 042700 KS) Price Chart



Date	Rating	Price (W)	Price Target (W)
18-Feb-25	OW	100900	150,000
31-Mar-25	OW	76500	100,000
30-Jun-25	OW	102900	125,000
23-Oct-25	N	150800	160,000
03-Nov-25	N	141500	150,000
07-Jan-26	UW	178100	150,000

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Feb 18, 2025. All share prices are as of market close on the previous business day.

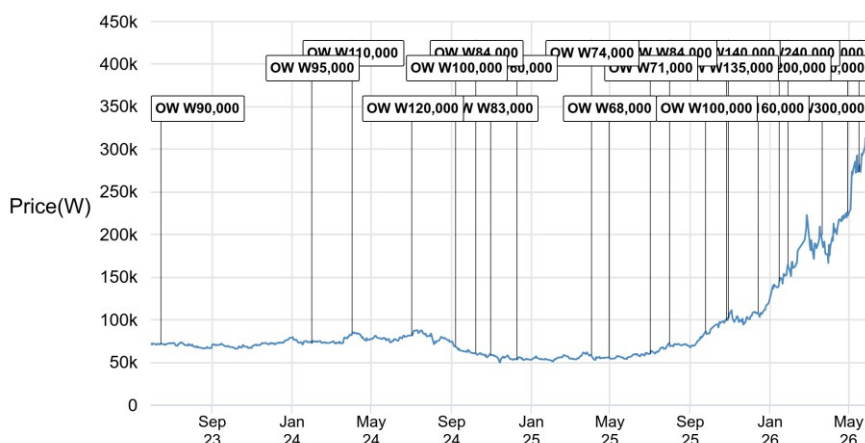
KIOXIA Holdings (285A) (285A.T, 285A JP) Price Chart



Date	Rating	Price (Y)	Price Target (Y)
23-Jan-26	OW	17910	25,000
22-Mar-26	OW	22360	38,000
17-May-26	OW	44450	80,000

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jan 23, 2026. All share prices are as of market close on the previous business day.

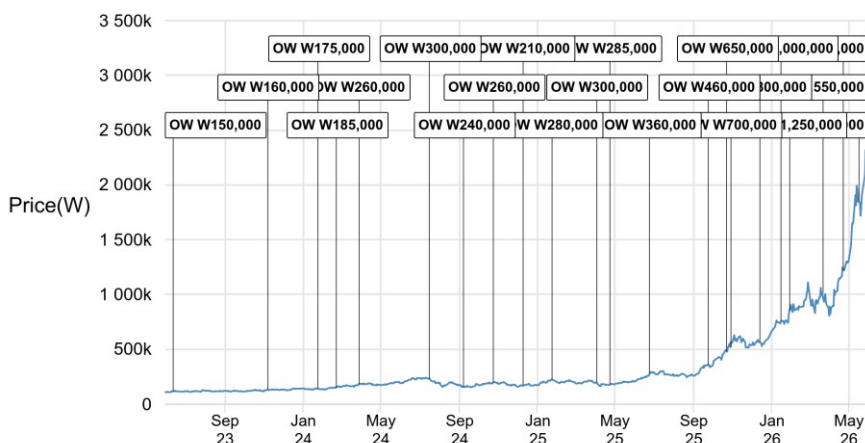
Samsung Electronics (005930.KS, 005930 KS) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Oct 09, 2011. All share prices are as of market close on the previous business day.

Date	Rating	Price (W)	Price Target (W)
15-Jun-23	OW	71900	90,000
01-Feb-24	OW	72700	95,000
03-Apr-24	OW	85000	110,000
03-Jul-24	OW	81800	120,000
08-Sep-24	OW	68900	100,000
08-Oct-24	OW	61000	84,000
31-Oct-24	OW	59100	83,000
10-Dec-24	N	53400	60,000
03-Apr-25	OW	58800	74,000
30-Apr-25	OW	55800	68,000
02-Jul-25	OW	60200	71,000
01-Aug-25	OW	71400	84,000
25-Sep-25	OW	85400	100,000
27-Oct-25	OW	98900	135,000
30-Oct-25	OW	101700	140,000
14-Dec-25	OW	108500	160,000
16-Jan-26	OW	145000	200,000
29-Jan-26	OW	164300	240,000
22-Mar-26	OW	199800	300,000
30-Apr-26	OW	226500	350,000
17-May-26	OW	273500	480,000

SK hynix (000660.KS, 000660 KS) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Dec 31, 2000. All share prices are as of market close on the previous business day.

Date	Rating	Price (W)	Price Target (W)
12-Jun-23	OW	115400	150,000
07-Nov-23	OW	133000	160,000
25-Jan-24	OW	141500	175,000
22-Feb-24	OW	149000	185,000
28-Mar-24	OW	181200	260,000
16-Jul-24	OW	230000	300,000
08-Sep-24	OW	156400	240,000
24-Oct-24	OW	196000	260,000
10-Dec-24	OW	168900	210,000
23-Jan-25	OW	225500	280,000
03-Apr-25	OW	197900	300,000
24-Apr-25	OW	181000	285,000
25-Jun-25	OW	278500	360,000
24-Sep-25	OW	361000	460,000
23-Oct-25	OW	481000	650,000
29-Oct-25	OW	519000	700,000
14-Dec-25	OW	571000	800,000
16-Jan-26	OW	756000	1,000,000
29-Jan-26	OW	862000	1,250,000
22-Mar-26	OW	1006000	1,550,000
23-Apr-26	OW	1220000	1,800,000
17-May-26	OW	1848000	3,000,000

Tokyo Electron (8035) (8035.T, 8035 JP) Price Chart



Date	Rating	Price (Y)	Price Target (Y)
27-Sep-23	OW	20165	24,000
09-Feb-24	OW	29435	34,200
09-Jul-24	OW	36040	41,000
11-Dec-24	OW	24600	35,000
13-Jun-25	OW	25000	31,000
12-Sep-25	OW	21405	26,000
08-Nov-25	OW	32800	38,000
10-Jan-26	OW	37910	45,000
13-Mar-26	OW	39750	51,000

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 04, 2001. All share prices are as of market close on the previous business day.

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